SignMuon, MuonSign, and the Role of Error Feedback

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Abstract

SignMuon (sign compression of the Muon update to one bit per parameter) is the most straightforward adaptation of Muon for training deep neural networks under an extremely low communication budget. Although SignMuon significantly outperforms SignSGD in practice, it may diverge even on a linear function. It is natural to conjecture that the order of operations is to blame, and that placing the sign before rather than after the Linear Minimization Oracle (LMO) might fix this issue. We show that the resulting algorithm, which we call MuonSign, still diverges even on a linear function. Thus, neither naive placement of the sign around the LMO yields a convergent algorithm. At the same time, EF21 does fix MuonSign: EF21-MuonSign provably achieves the standard $\mathcal{O}(T^{-1/2})$ rate for smooth nonconvex optimization, decisively outperforms SignSGD, and matches SignMuon when training various deep learning models – on image classification, in federated settings, and on language modelling – providing a reliable drop-in replacement for SignSGD. Compressing the downlink as well is where the guarantee starts to cost something, and we measure what.

1 Introduction

Modern deep neural network (DNN) training methods require optimizers that not only achieve fast convergence but also minimize communication overhead in distributed systems. In federated learning, data are distributed across clients that communicate with a central server to train a global model. Each client possesses its own dataset and updates its parameters using a local optimizer. The goal of federated learning is to minimize the average loss function across all clients. Under limited network bandwidth, transmitting full model updates can significantly slow down the training process. Gradient compression methods are commonly used to address this issue (Bernstein et al. 2018, 2019; Beznosikov et al. 2024).

For instance, the SignSGD algorithm has demonstrated that even extreme compression (down to 1 bit per component) can preserve the practical effectiveness of stochastic optimization while reducing the communication volume. This data volume is reduced by a factor of approximately $32\times$ compared to standard SGD, with only minor degradation in model quality (Bernstein et al. 2018, 2019). However, SignSGD does not account for the matrix structure of the parameters. In tasks with matrix weights, it treats matrices as one-dimensional vectors (flattening), ignoring their intrinsic two-dimensional tensor structure, which may negatively affect the training performance of DNNs.

Recently, Muon has demonstrated that orthonormalized updates can stabilize DNN training and, in several cases, outperform standard adaptive optimization methods in terms of final solution accuracy (Jordan et al. 2024; Bernstein and Newhouse 2024; Essential AI 2025). For matrix parameters, Muon implements a steepest descent step in the spectral norm. According to Jordan et al. (2024), the update direction is computed as the orthonormal matrix $\mathbf{U}\mathbf{V}^\top$, where $\mathbf{U}$ and $\mathbf{V}$ are the singular-vector matrices obtained from the gradient decomposition. Based on the structure of the algorithm, Muon requires the transmission of full update matrices, which can be prohibitively computationally expensive for federated tasks.

We study the natural ways to combine sign compression with the Muon LMO while communicating only one bit per parameter: **SignMuon**, which places the sign operator *after* the LMO and computes $\mathrm{sign}(\mathrm{LMO}(\cdot))$; **MuonUSign**, which places it *before* and computes $\mathrm{LMO}(\mathrm{sign}(\cdot))$; and **MuonSign**, which signs on *both* sides, $\mathrm{sign}(\mathrm{LMO}(\mathrm{sign}(\cdot)))$, compressing the uplink and the downlink alike. All three preserve the matrix-aware geometry of Muon and, empirically, match Muon's accuracy while significantly outperforming SignSGD in centralized and federated learning.

However, none of these combinations enjoys a global convergence guarantee. We prove that all three can diverge: SignMuon on a 4×4 counterexample (Theorem 1), and MuonUSign and MuonSign on a single common 5×5 counterexample (Theorems 2–3); in each case the 1-bit step becomes an ascent direction on a linear objective. To restore convergence without increasing the communication volume, we introduce **EF21-MuonUSign**, an Error-Feedback variant adapted from EF21-Muon Gruntkowska et al. (2025): the scaled-sign operator compresses only the residual between the momentum and a gradient estimator, while the descent step uses the full Muon LMO of the reconstructed estimator on the server. We further introduce **EF21-MuonSign**, which adds a second error-feedback loop on the downlink to compress both channels to one bit per parameter. Both converge on the counterexamples above while retaining one-bit communication.

Our experiments investigate how effectively Muon-style structured update directions can be combined with the extreme communication compression characteristic of sign-based methods. Specifically, we analyze how transmitting only the update direction signal affects the accuracy and stability of training compared to using the full Muon update. We validate SignMuon and the Error-Feedback variants on centralized image classification on CIFAR-10 with a ResNet-18 backbone, federated image classification on CIFAR-10 with a CNN across several client counts, and on a synthetic smooth convex least-squares problem. Our contributions are:
(i) SignMuon, MuonUSign, and MuonSign, matrix-aware 1-bit optimizers that place the sign operator after, before, and on both sides of the LMO, respectively;
(ii) counterexamples (Theorems 1–3) proving that all three may diverge;
(iii) EF21-MuonUSign, which compresses the uplink to one bit per parameter while restoring convergence guarantees;
(iv) EF21-MuonSign, a bidirectional variant that adds a downlink error-feedback loop, keeping both channels at one bit per parameter.

2 Related Work

SignSGD. Vanilla SignSGD, introduced by Bernstein et al. (2018) and later applied to federated learning by Bernstein et al. (2019), required increasing batch sizes to guarantee convergence. This issue, together with SignSGD's inherent bias, was addressed through the adoption of the Error Feedback technique (Karimireddy et al. 2019). An alternative approach is to randomize the sign operator itself (Chen et al. 2020; Safaryan and Richtarik 2021; Jin et al. 2025).

Compression for Muon. Gruntkowska et al. (2025) developed a theoretical framework for federated learning with error feedback in which Muon/Gluon (Riabinin et al. 2025) updates are compressed bidirectionally (client-to-server and server-to-client). Qian et al. (2026) employed variance reduction to lower the compression error for Gluon under unbiased and contractive compressors. Concurrently with our work, Mishra, Trivedi, and Kumar (2026) proposed vanilla SignMuon and demonstrated its empirical feasibility, and stated an $\mathcal{O}(1/\sqrt{T})$ convergence guarantee for it; in Appendix A.5 we show that their finalized rate in fact covers only the gradient-sign (signSGD) instantiation, while the bound for the actual polar-sign method is left uninformative on exactly the instances of Theorem 1.

3 Problem Statement

Centralized Learning: We consider the stochastic optimization problem:

$$\min_{\mathbf{X}\in\mathcal{X}} \{f(\mathbf{X}) := \mathbb{E}_{\xi\sim\mathcal{D}}[f_\xi(\mathbf{X})]\}, \tag{1}$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m\times n}$), functions $f_\xi : \mathcal{X} \to \mathbb{R}$ are continuously differentiable (possibly non-convex). Here, $f_\xi(\mathbf{X})$ is the loss function of a model parameterized by $\mathbf{X}$, computed at a data point ξ, sampled from the probability distribution $\mathcal{D}$.

Federated Learning: In modern machine learning, progress is increasingly driven by the ability to scale computational systems. SOTA models rely on massive datasets and highly complex architectures, often requiring substantial computational resources and extended training times. As a result, the training process is formulated as an optimization problem in a distributed environment:

$$\min_{\mathbf{X}\in\mathcal{X}} \left\{ f(\mathbf{X}) := \frac{1}{N}\sum_{j=1}^{N} f_j(\mathbf{X}) \right\}, \qquad f_j(\mathbf{X}) = \mathbb{E}_{\xi_j\sim\mathcal{D}_j}[f_j(\mathbf{X};\xi_j)] \tag{2}$$

where $\mathbf{X}\in\mathcal{X}$ are the model parameters, $N\geq 1$ is the number of clients, and $f_j(\mathbf{X})$ is the loss of the model $\mathbf{X}$ on the data $\mathcal{D}_j$, stored on client $j\in[N] := \{1,\ldots,N\}$. In federated learning, clients do local computations and synchronize with a global server. Since communication bandwidth is often limited, the volume of transmitted data becomes a major bottleneck in the training process (Bernstein et al. 2019; Horváth et al. 2019). Several practical techniques for federated optimization and communication compression have also been studied in (Beznosikov et al. 2024; Gruntkowska et al. 2025; Ahn et al. 2025).

For theoretical optimization analysis in distributed environments, we rely on Assumption 2 that the objective function f is smooth w.r.t. the spectral norm. We note that, in practice, weaker smoothness assumptions such as the generalized (L_0, L_1)-smoothness condition may also be considered (Pethick et al. 2026). Limited network bandwidth makes the transmission of full-precision 32-bit gradients prohibitively expensive, especially when training large-scale models. This motivates the study of communication-efficient optimization methods based on extreme gradient compression (Bernstein et al. 2018, 2019; Beznosikov et al. 2024).

Linear Minimization Oracle (LMO) is defined as an operator $A(\cdot)$ that provides the solution to a linear minimization problem over a feasible set. In particular, the constraints can be defined via the unit ball of a given norm:

$$A(\mathbf{X}) = \arg\min_{\mathbf{Y}\in\{\mathbf{Y}\in\mathcal{S}\,|\,\|\mathbf{Y}\|\leq 1\}} \langle \mathbf{X}, \mathbf{Y}\rangle, \tag{3}$$

where $\mathbf{X}\in\mathbb{R}^{m\times n}$ is the input matrix, $\mathcal{S}\subset\mathbb{R}^{m\times n}$ is a convex set, and $\langle \mathbf{X}, \mathbf{Y}\rangle = \sum_{i,j} X_{ij}Y_{ij}$ is the standard matrix inner product (Pethick et al. 2025).

Such an operator selects the direction of steepest descent with respect to the specified norm. The analytical and practical role of the LMO is discussed in detail in the literature on norm-constrained LMO optimizers (Pethick et al. 2025; Kovalev 2025; Riabinin et al. 2025).

Muon optimizer uses the matrix structure of the gradient. Muon implements a steepest descent step in the spectral norm, where the update direction is obtained by orthogonalizing the gradient matrix (Bernstein and Newhouse 2024). Let $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition

(SVD) of the matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, where $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ are the orthonormal matrices of singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$ is the diagonal matrix of singular values, and $r = \mathrm{rank}(\mathbf{M})$. Then, the LMO direction takes the following form:

$$A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top \in \mathbb{R}^{m\times n}. \tag{4}$$

That is, Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. In practice, Muon utilizes a smoothed gradient estimate with momentum, and its update is defined as follows:

$$\begin{aligned} \mathbf{M}_t &= \mu\mathbf{M}_{t-1} + \mathbf{G}_t, \\ \mathbf{M}_t &= \mathbf{U}_t\mathbf{\Sigma}_t\mathbf{V}_t^\top, \\ \mathbf{D}_t &= -A(\mathbf{M}_t) = \mathbf{U}_t\mathbf{V}_t^\top \\ \mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t\mathbf{D}_t. \end{aligned} \tag{5}$$

In the original Muon implementation, the computation of $\mathbf{U}_t\mathbf{V}_t^\top$ is approximated using Newton-Schulz iterations and related polynomial schemes. It enables efficient matrix orthogonalization without an explicit SVD computation (Amsel et al. 2026; Li and Hong 2025; Liu et al. 2025). Such orthogonalization produces more isotropic updates and prevents a small number of dominant singular directions from controlling the optimization process, which is particularly beneficial for training DNN (Bernstein and Newhouse 2024; Essential AI 2025).

Despite its advantages, Muon requires transmitting the full update matrix $\mathbf{D}_t$, making it expensive in federated settings. Our objective is to combine Muon's matrix-aware geometry with extreme (1-bit) communication compression. SignMuon, MuonUSign, and MuonSign achieve this combination empirically, but, as we show (Theorems 1–3), they lack global convergence guarantees. The Error-Feedback variants restore convergence: in Appendix A.7 we show that EF21-MuonUSign and EF21-MuonSign are *exact instances* of the EF21-Muon framework of Gruntkowska et al. (2025), so that its guarantees transfer without any new analysis and yield the standard $\mathcal{O}(T^{-1/2})$ rate for smooth non-convex optimization (Theorem 5):

$$\min_{0\le t\le T} \mathbb{E}\big[\|\nabla f(\mathbf{X}_t)\|_*^2\big] = \mathcal{O}(T^{-1/2}), \tag{6}$$

as well as an $\mathcal{O}(T^{-1/4})$ rate for $\min_t \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*$ under the weaker layer-wise (L^0, L^1)-smoothness with a plain constant learning rate (Corollary 2). Finally, EF21-MuonSign adds a second error-feedback loop on the downlink, making communication 1-bit in *both* directions. The $\mathcal{O}(T^{-1/2})$ rate survives, but not for free: a sign-based downlink is contractive only in the Euclidean norm, whereas Muon's spectral geometry asks for contractivity in the layer norm, so the admissible step size shrinks by a further factor of the layer rank. We show that this gap cannot be closed for the scaled sign by any sharper analysis – only by changing the compressor (Remark 7) – and we measure what it costs on a transformer (Section 5.4).

4 Theory

Consider the **SignA** family of algorithms, where A denotes any optimizer based on LMO directions and Sign is sign-compressor. All methods in this family follow the same three-stage structure: momentum accumulation, computation of an LMO direction, and sign-compression of the resulting update.

Formally, a SignA method initializes the momentum buffer as $\mathbf{M}_0 = \mathbf{0}$, and at each iteration $t \ge 1$ performs the following sequence of updates:

$$\begin{aligned} \mathbf{M}_t &= \mu\mathbf{M}_{t-1} + \mathbf{G}_t && \text{(Momentum accumulation)} \\ \tilde{\mathbf{M}}_t &= \begin{cases} \mathbf{G}_t + \mu\mathbf{M}_t & \text{(Nesterov momentum)}, \\ \mathbf{M}_t & \text{(otherwise)} \end{cases} && \text{(Effective direction)}, \\ \mathbf{D}_t &= -A(\tilde{\mathbf{M}}_t), && \text{(LMO direction)} \\ \mathbf{s}_t &= \mathrm{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m\times n}, && \text{(Sign compression)} \\ \mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t\mathbf{s}_t && \text{(Parameter update)}, \end{aligned} \tag{7}$$

where $\mathbf{X}_t \in \mathbb{R}^{m\times n}$ is the current model parameter matrix, $\mathbf{G}_t$ is its stochastic gradient, $\mu \in [0, 1)$ is the momentum coefficient and η_t is the learning rate. The operator $A(\cdot)$ denotes a Linear Minimization Oracle (LMO) in a specified non-Euclidean norm, and the function $\mathrm{sign}(\cdot)$ is applied element-wise to the resulting structured direction $\mathbf{D}_t$.

The proposed SignMuon algorithm is a particular instance of the SignA family. It combines the orthogonalized LMO directions of the Muon optimizer with the extreme sign-based compression strategy of SignSGD. MuonUSign is the mirror image of SignMuon: it applies the sign operator *before* the LMO. Starting from the effective momentum direction $\tilde{\mathbf{M}}_t$, it compresses it to $\mathrm{sign}(\tilde{\mathbf{M}}_t)$ and then takes the Muon LMO of that sign matrix, stepping along the resulting orthonormal direction (Algorithm 4):

$$\mathbf{D}_t = -A\big(\mathrm{sign}(\tilde{\mathbf{M}}_t)\big), \qquad \mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t\,\mathbf{D}_t. \tag{8}$$

MuonSign (Algorithm 5) signs both sides, additionally compressing the LMO output to $\mathrm{sign}(\mathbf{D}_t)$, so that both the uplink and the downlink cost one bit per parameter. In the following sections, we present SignMuon, MuonUSign, and MuonSign, each in centralized and federated variants.

4.1 Centralized Learning

In the centralized setting, SignMuon is applied to the matrix-valued parameters (weights of convolutional and fully connected layers, $n_{dim} \ge 2$), while vector parameters (biases, BatchNorm, $n_{dim} = 1$) and the final classification layer are optimized with AdamW – the design introduced by Jordan et al. (2024) and formalized in (Li and Hong 2025; Riabinin et al. 2025).

For matrix parameters, the key component of SignMuon is the computation of an orthonormal LMO direction $\mathbf{D}_t \approx \mathbf{U}_t\mathbf{V}_t^\top$. To avoid the computational cost of performing a full singular value decomposition (SVD) at every optimization step, the orthogonalization is approximated using a 5th-order Newton-Schulz iteration (Algorithm 1). In our implementation, the Newton-Schulz coefficients are chosen as $a = 3.4445, b = 4.7750, c = 2.0315$. The main difference between SignMuon and the classical SignSGD algorithm lies in the object being compressed. In SignSGD, sign compression is applied directly to the stochastic gradient. In contrast, SignMuon first constructs a structured update direction $\mathbf{D}_t$, corresponding to the Muon LMO step and only then applies the elementwise sign operator to obtain $\mathrm{sign}(\mathbf{D}_t)$. Consequently, SignMuon preserves the geometry of orthonormal updates while reducing the transmitted information to one bit per component (Algorithm 2 in Appendix A.15); MuonSign is built from the same idea (Algorithm 5). One further implementation choice is not free: the two sign placements produce step matrices whose norms scale differently with the shape of the layer, so a single global step size cannot be correct for both. We therefore set the step size of each layer by the *unit-gain* rule derived in Appendix A.14, which fixes the shape dependence a priori and leaves a single shape-free base rate η_0 to be tuned.

4.2 Theoretical limitations

A natural hope is that the Muon LMO direction—the steepest-descent direction in the spectral norm—remains a descent direction after sign compression, whether the sign is applied before or after the oracle. We show that this hope fails for all three one-bit placements of the sign operator around the LMO: in each case the compressed step can become an *ascent* direction on a smooth objective, so none of the methods admits a global convergence guarantee.

Write $\mathrm{polar}(\mathbf{Y}) := -A(\mathbf{Y}) = \mathbf{U}\mathbf{V}^\top$ for the Muon LMO direction, i.e. the orthogonal polar factor read off the (rank-truncated) SVD $\mathbf{Y} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. The three methods share the update $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t\,\mathbf{s}_t$ and differ only in where the sign acts on the effective direction $\tilde{\mathbf{M}}_t$:

$$\mathbf{s}_t = \begin{cases} \mathrm{sign}\big(\mathrm{polar}(\tilde{\mathbf{M}}_t)\big) & \text{(SignMuon: after)},\\ \mathrm{polar}\big(\mathrm{sign}(\tilde{\mathbf{M}}_t)\big) & \text{(MuonUSign: before)},\\ \mathrm{sign}\big(\mathrm{polar}(\mathrm{sign}(\tilde{\mathbf{M}}_t))\big) & \text{(MuonSign: both)}. \end{cases} \tag{9}$$

SignMuon and MuonUSign transmit one bit per parameter on the uplink; MuonSign compresses the downlink as well, so both channels cost one bit per parameter.

We refute the descent property on *linear* objectives—the simplest smooth functions, on which any sound method must make unbounded progress:

$$f(\mathbf{W}) = \langle \mathbf{G}, \mathbf{W}\rangle = \mathrm{Tr}(\mathbf{G}^\top \mathbf{W}), \qquad \nabla f(\mathbf{W}) \equiv \mathbf{G}. \tag{10}$$

Gradient descent and full-precision Muon drive $f \to -\infty$ on (10); a step rule that instead drives $f \to +\infty$ is unambiguously ascending. The following elementary reduction collapses each method to a single scalar inequality and removes momentum from the discussion entirely.

Proposition 1 (Reduction on linear objectives) *Run any of the three methods* (9) *on the linear objective* (10) *with* $\mathbf{G} \neq \mathbf{0}$*, from an arbitrary* $\mathbf{X}_0$*, with any momentum coefficient* $\mu \in [0, 1)$ *under either the Standard or the Nesterov rule. Then* $\tilde{\mathbf{M}}_t = \gamma_t\,\mathbf{G}$ *with* $\gamma_t > 0$*, the update direction is the constant matrix* $\mathbf{s}(\mathbf{G})$ *obtained by substituting* $\mathbf{G}$ *for* $\tilde{\mathbf{M}}_t$ *in* (9)*, and*

$$f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = -\eta_t\,\big\langle \mathbf{G},\ \mathbf{s}(\mathbf{G})\big\rangle. \tag{11}$$

In particular, if $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$ *then* f *strictly increases at every iteration for any* $\eta_t > 0$*, and* $f(\mathbf{X}_t) \to +\infty$ *whenever* $\sum_t \eta_t = \infty$ *(e.g. for any constant step size).*

The proof (Appendix A.1) is one line: the momentum buffer $\mathbf{M}_t = \big(\sum_{i=0}^{t-1} \mu^i\big)\mathbf{G}$ is a positive multiple of $\mathbf{G}$, hence so is $\tilde{\mathbf{M}}_t$ under both rules, and both $\mathrm{sign}(\cdot)$ and $\mathrm{polar}(\cdot)$ are invariant under multiplication by a positive scalar. It therefore suffices, for each method, to exhibit a single gradient $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$; momentum is irrelevant.

SignMuon (sign after the LMO). The step follows the *sign* of the Muon direction, so the descent condition is $\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G}))\rangle > 0$, and $\mathrm{sign}(\mathrm{polar}(\mathbf{G}))$ need not correlate with $\mathbf{G}$.

Theorem 1 (Divergence of SignMuon) *There is a matrix* $\mathbf{G} \in \mathbb{R}^{4\times 4}$ *with* $\big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G}))\big\rangle = -\frac{42468}{103} < 0$*. Hence SignMuon ascends on* (10)*:* f *strictly increases at every iteration for all* $\eta_t > 0$*, all* $\mu \in [0, 1)$*, and both momentum variants.*

The construction (Appendix A.2) takes $\mathbf{G} = 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}$ with $\mathbf{O}$ orthogonal, so that $\mathrm{polar}(\mathbf{G}) = \mathbf{O}$ while the sign pattern of $\mathbf{O}$ is anticorrelated with the dominant component $\mathbf{u}_1\mathbf{v}_1^\top$.

MuonUSign (sign before the LMO). Here the step is the *full* orthonormal direction $\mathrm{polar}(\mathrm{sign}(\mathbf{G}))$, so one might hope the LMO "heals" the coarseness of the sign; the descent condition is $\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\rangle > 0$. It fails, because applying the spectral LMO to a sign matrix can rotate the update away from $\mathbf{G}$.

Theorem 2 (Divergence of MuonUSign) *There is a matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *with* $\big\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\big\rangle \approx -13.89 < 0$*. Hence MuonUSign ascends on* (10) *for all* $\eta_t > 0$*, all* $\mu \in [0, 1)$*, and both momentum variants.*

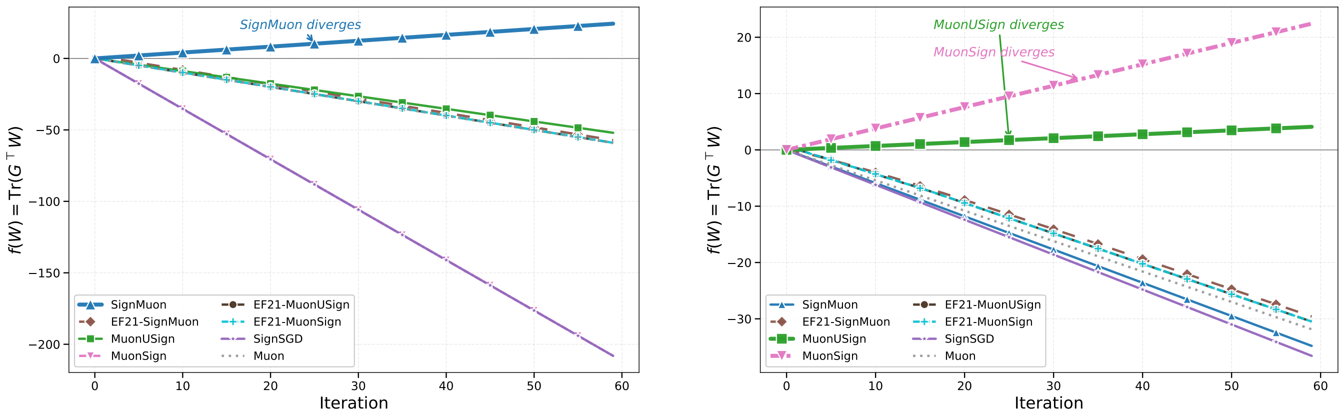
MuonSign (sign before and after the LMO). Signing the LMO output as well only discards information, so the bidirectional method inherits the same failure; with $\mathbf{s}(\mathbf{G}) = \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G})))$ the descent condition is $\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G})))\rangle > 0$.

Theorem 3 (Divergence of MuonSign) *For the same matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *as in Theorem 2,* $\big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G})))\big\rangle = -76 < 0$*. Hence MuonSign ascends on* (10) *for all* $\eta_t > 0$*, all* $\mu \in [0, 1)$*, and both momentum variants.*

The proofs are in Appendices A.2–A.4, and Figure 1 shows the three ascending trajectories against the descending Muon, SGD, and EF21 baselines. Together, Theorems 1–3 show that no placement of the sign operator around the Muon LMO yields a convergent method, which motivates the Error-Feedback variants introduced next.

4.3 Centralized Error Feedback: EF21-SignMuon

Theorems 1–3 rule out every static placement of the sign around the LMO. The classical remedy for a biased one-bit compressor is *error feedback*: store the compression



(a) SignMuon (4×4, Theorem 1)

(b) MuonUSign and MuonSign (5×5, Theorems 2–3)

Figure 1: Optimization trajectories on the linear objective $f(\mathbf{W}) = \mathrm{Tr}(\mathbf{G}^\top \mathbf{W})$ of Theorems 1–3. **(a)** SignMuon strictly ascends on the 4×4 instance; **(b)** MuonUSign and MuonSign both strictly ascend on the single common 5×5 instance. In both panels the full-precision Muon, SGD, and Error-Feedback (EF21-MuonUSign, EF21-MuonSign) baselines minimize the objective. Trajectories are shown momentum-free ($\mu = 0$); since the gradient is constant and the Muon LMO is scale-invariant, by Proposition 1 the trajectory is identical for every $\mu \in [0, 1)$ and both momentum variants, so momentum has no effect here.

error and reinject it at the next step rather than discarding it. Error feedback has long been known to repair sign-based methods—it fixes SignSGD and related gradient-compression schemes (Karimireddy et al. 2019)—and was later simplified and sharpened into the EF21 mechanism (Richtárik, Sokolov, and Fatkhullin 2021). Its most direct use for SignMuon keeps the geometry—the sign *after* the LMO—and error-feedbacks the oracle's output. The resulting method, **EF21-SignMuon** (Algorithm 3), tracks an estimate $\mathbf{d}_t^{\mathrm{est}}$ of the LMO direction $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$, refreshed by a scaled sign of the residual,

$$\begin{aligned} \mathbf{d}_t^{\mathrm{est}} &= \mathbf{d}_{t-1}^{\mathrm{est}} + \alpha_t \, \mathrm{sign}\big(\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big), \\ \alpha_t &= \mathrm{mean}\big|\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big|, \end{aligned} \tag{12}$$

and steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \, \mathbf{d}_t^{\mathrm{est}}$.

Error feedback on the compressed *output*, however, does not restore the descent property—and it fails *universally*, at every problem scale and momentum setting.

Theorem 4 (Divergence of EF21-SignMuon) *For every $L > 0$, step size $\eta > 0$, momentum coefficient $\mu \in [0, 1)$, and either momentum variant, there is an L-smooth (Assumption 2), bounded-below (Assumption 1) function $f : \mathbb{R}^{2\times 2} \to \mathbb{R}$ on which EF21-SignMuon started at $\mathbf{X}_0 = \mathbf{0}$ diverges: for an explicit constant $c = c(f) > 0$,*

$$f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = c\, L\eta^2 > 0 \qquad (t \geq 3), \tag{13}$$

so $f(\mathbf{X}_t) \to +\infty$. In particular, no step-size rule $\eta = \eta(L, \mu)$ using only the smoothness and momentum constants can make the method convergent.

The obstruction (proof in Appendix A.6) is specific to signing the LMO output: a *single* magnitude α_t rescales all entries' signs at once, so a large, sign-flipping off-diagonal pins $\alpha_t = \Theta(1)$ and swamps the small diagonal target, whose estimate then locks onto the *wrong* sign and drives one coordinate of $\mathbf{X}_t$ to infinity. This is not a degeneracy—the matrices involved have condition number $\frac{16}{9}$—and momentum does not help: Figures 4–5 (Appendix A.6) show EF21-SignMuon as the lone ascending method, at the exact rate $\frac{49}{480}$ per step for every μ and both momentum variants, while all other methods stay bounded. The remedy is to error-feedback the gradient *before* the LMO rather than its output—the subject of the next subsection.

4.4 Centralized Error Feedback: EF21-MuonUSign and EF21-MuonSign

Theorem 4 pinpoints the defect of EF21-SignMuon: coupling all coordinates through a single magnitude α_t is fatal only because the sign sees the LMO *output*. Compressing *before* the oracle removes the coupling—the LMO then acts on a faithfully de-biased matrix—and, as we show next, restores convergence at the same one-bit uplink cost. Concretely, we replace the sign-compressed step $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \, \mathrm{sign}(\mathbf{D}_t)$ by an error-feedback variant that adapts the EF21-Muon mechanism (Gruntkowska et al. 2025) to sign compression.

We define the **USign** compressor as the directional pair $\mathrm{USign} := (\mathcal{C}_\uparrow, \mathcal{C}_\downarrow)$, where $\mathcal{C}_\uparrow(\mathbf{Y}) = \mathrm{mean}\,|\mathbf{Y}|\,\mathrm{sign}(\mathbf{Y})$ is the scaled-sign operator on the uplink (client $\to$ server) and $\mathcal{C}_\downarrow = I$ is the identity on the downlink (server $\to$ client). Unlike a fully 1-bit scheme, USign compresses only the uplink to one bit per parameter, while the server broadcasts the full model. **EF21-MuonUSign** pairs USign with EF21 and the Muon LMO: the sign acts on the EF21 residual, and the optimization step uses the full Muon LMO of the reconstructed estimator.

EF21-MuonUSign maintains a gradient estimator $\mathbf{g}_t^{\mathrm{est}}$ updated via a contractive sign compressor on the residual $\Delta_t = \mathbf{M}_t - \mathbf{g}_t^{\mathrm{est}}$:

$$\alpha_t = \mathrm{mean}(|\Delta_t|), \quad \mathbf{g}_{t+1}^{\mathrm{est}} = \mathbf{g}_t^{\mathrm{est}} + \alpha_t \, \mathrm{sign}(\Delta_t), \tag{14}$$

and takes the descent step

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t\ \mathbf{D}_t,$$
$$\mathbf{D}_t = -A(\mathbf{g}_{t+1}^{\text{est}}) \approx \mathbf{U}_{t+1}\mathbf{V}_{t+1}^\top, \quad (15)$$

As $\mathbf{g}_t^{\text{est}} \to \mathbf{M}_t$, the direction $\mathbf{D}_t \to \mathbf{U}_t\mathbf{V}_t^\top$ is always a descent direction, since $\langle \nabla f,\, \mathbf{U}_t\mathbf{V}_t^\top \rangle \approx \|\nabla f\|_* > 0$. The sign in (14) acts only on the internal compression residual, keeping the uplink at one bit per parameter. The Algorithm 6 is given in AppendixA.15. Figure 1 confirms that EF21-MuonUSign converges on the same counterexample where SignMuon diverges.

While USign leaves the downlink uncompressed, a fully 1-bit scheme is obtained by compressing both channels. We define the **UDSign** compressor as the directional pair $\mathrm{UDSign} := (\mathcal{C}_\uparrow,\, \mathcal{C}_\downarrow)$ with the same scaled-sign operator on both channels, $\mathcal{C}_\uparrow(\mathbf{Y}) = \mathcal{C}_\downarrow(\mathbf{Y}) = \mathrm{mean}\,|\mathbf{Y}|\,\mathrm{sign}(\mathbf{Y})$. In this version of Error Feedback Algorithm the node keeps the exact model $\mathbf{X}_t$ and takes the exact LMO step, while it observes only a compressed model shift $\mathbf{W}_t$ reconstructed by error feedback,

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{D}_t,$$
$$\mathbf{W}_{t+1} = \mathbf{W}_t + \alpha_t^{\downarrow}\,\mathrm{sign}(\mathbf{X}_{t+1} - \mathbf{W}_t), \quad (16)$$

with the gradient evaluated at $\mathbf{W}_t$ and $\mathbf{D}_t = -A(\mathbf{g}_{t+1}^{\text{est}})$. The full procedure is in Algorithm 7.

It should be noted that both error feedback methods require additional memory: each compressed channel maintains its own error feedback buffer (the gradient estimate $\mathbf{g}^{\text{est}}$ and the model shift buffer $\mathbf{W}$ in the case of bidirectional feedback). This additional overhead is low in terms of server memory, but may be unwanted on worker nodes with limited memory.

4.5 Federated Learning

In federated learning, the training data are distributed across N clients. Each client j possesses a local dataset $\mathcal{D}_j$ and computes a local loss function $f_j(\mathbf{X})$. The goal of the training process is to minimize the averaged loss function across all clients (2). Since clients do not exchange raw data and communicate only through messages sent to the server, communication efficiency becomes a critical concern in federated optimization.

In the federated version of SignMuon, the server acts as an aggregation node, while the main optimization procedure is performed locally on the clients. At the beginning of communication round t, each client j receives a copy of the current global model. Rather than performing local parameter updates, the client computes an averaged stochastic gradient by accumulating gradients over $\mathrm{n\_steps}$ local mini-batches. This gradient accumulation procedure reduces the variance of the stochastic gradient estimate before the application of the LMO operator. Formally, the accumulated gradient is defined as:

$$\mathbf{G}_t^{(j)} \leftarrow \frac{1}{n_{\text{steps}}} \sum_{i=1}^{n_{\text{steps}}} \nabla f_j(\mathbf{X}_t; \xi_{t,i}^{(j)}).$$

In accordance with the general SignA framework (7), each client maintains its own local momentum buffer $\mathbf{M}_t^{(j)}$. After computing the accumulated gradient, the client updates the momentum estimate, applies the LMO (implemented via Newton-Schulz orthogonalization, Algorithm 1), and then performs element-wise sign compression:

$$\mathbf{M}_t^{(j)} = \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$$

$$\mathbf{s}_t^{(j)} = \mathrm{sign}\big(A(\mathbf{M}_t^{(j)})\big) \in \{\pm 1\}^{m \times n}$$

where $t \in [0, \mathrm{rounds} - 1]$, $A(\cdot)$ is the Muon LMO (Newton-Schulz, as in the centralized setting). Thus, each client transmits only 1 bit per matrix parameter component to the server.

On the server, messages received from the clients are aggregated using a majority vote. The aggregated sign is denoted by $\mathbf{s}_t^{\text{agg}}$, then it is given by the formula:

$$\mathbf{s}_t^{\text{agg}} = \mathrm{sign}\left(\sum_{j=1}^{N} \mathbf{s}_t^{(j)} \right). \quad (17)$$

The majority-vote operation ensures that the resulting sign $\mathbf{s}_t^{\text{agg}}$ is equal to $+1$ or -1 in each component and corresponds to the “majority vote” of the clients. Since momentum has already been taken into account at the clients’ level, the server directly updates the matrix parameters of the global model:

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{s}_t^{\text{agg}}.$$

A special treatment is applied to the final classification layer of the model. These parameters are not updated using the SignMuon rule; instead, they are optimized using AdamW. The complete federated SignMuon procedure is presented in AppendixA.15.

4.6 Federated Learning with Error Feedback

Although Federated SignMuon demonstrates strong empirical stability, sign-based compression inevitably introduces bias into the descent direction. To compensate for quantization error and ensure strong convergence guarantees, we incorporate an Error Feedback method. In the federated setting, the Error-Feedback mechanism is adapted from EF21-Muon (Gruntkowska et al. 2025).

The LMO orthogonalization is moved from the clients to the server: each client evaluates the difference between its local momentum $\mathbf{M}_t^{(j)}$ and the previous gradient estimate $\mathbf{g}_t^{(j)}$, compresses it via scaled sign, and sends only $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$. The server aggregates these, updates a global gradient estimator $\mathbf{G}_{t+1}$, and applies the Muon LMO. The complete Algorithm 9 is in AppendixA.15. The transmission of an additional scalar $\alpha_t^{(j)}$ for each layer introduces insignificant communication overhead, keeping the uplink communication cost at $\approx$ 1 bit per parameter (the server broadcasts the full model on the downlink). Thus, all federated variants – SignMuon, EF21-MuonUSign and EF21-MuonSign – effectively address: the baseline majority-vote scheme and the error-feedback variant – effectively address the communication bottleneck in federated learning. They preserve the geometric benefits of Muon-style matrix orthogonalization while maintaining an extremely low communication budget.

Algorithm	Learning rate (η)	Momentum (μ)	Iterations to 0.001
SignMuon	0.0002	0.2	2104
MuonUSign	-	-	-
EF21-MuonUSign	0.0033	0.1	3694
EF21-MuonSign	0.0028	0.1	3564
Muon	0.0065	0.1	1122
SignSGD	0.00015	0.8	3120
SGD	0.1	0.95	972
Adam	0.07	-	202

Table 1: Tuned learning rates and momentum coefficients in the experiments on the smooth convex optimization problem (Equation 18).

5 Experiments

The goal of these experiments is empirical validation of the SignMuon, EF21-MuonUSign and EF21-MuonSign algorithms. The study consists of several parts, including evaluation in both centralized and federated settings, as well as validation on a synthetic least-squares optimization problem.

5.1 Smooth Convex Optimization Problem (Synthetic Experiment)

To evaluate the effect of matrix structure on the optimization process (without the influence of stochastic noise or the complexity of DNN architectures), we begin our experiments with a deterministic L-smooth convex optimization problem. We consider the minimization of a quadratic form:

$$F(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}^{1/2}\mathbf{X}\mathbf{B}^{1/2}\|_F^2 = \frac{1}{2}\langle \mathbf{X}, \mathbf{A}\mathbf{X}\mathbf{B}\rangle \to \min_{\mathbf{X}\in\mathbb{R}^{m\times n}}, \quad (18)$$

where the parameter matrix $\mathbf{X} \in \mathbb{R}^{m\times n}$ has dimension $m = n = 500$, which is representative of the size of parameter matrices in deep neural networks. The matrices $\mathbf{A} \in \mathbb{S}_+^m$ and $\mathbf{B} \in \mathbb{S}_+^n$ are symmetric positive semidefinite. To generate the problem instance, we randomly construct the spectra of $\mathbf{A}$ and $\mathbf{B}$ by sampling their eigenvalues uniformly from the interval $(0, 1)$, which guarantees a smoothness constant $L \leq 1$ with respect to the Frobenius norm. The parameter matrix $\mathbf{X}_0$ is initialized with elements sampled independently from a normal distribution $\mathcal{N}(0, 0.01)$.

Since theoretical convergence guarantees depend on the choice of norm and the corresponding smoothness constants, we adopt an empirical evaluation criterion. Specifically, the optimizers are compared by the minimum number of iterations required to reach the target function $F(\mathbf{X}) \leq 0.001$, with a maximum number of iterations set to $T_{\max} = 5000$. We perform a grid search over the hyperparameter ranges for each of the optimization algorithms under consideration (Table 5).

The results of the grid search with the number of iterations required to converge to 0.001, are summarized in Table 1. Muon converges in 1122 iterations, comparable to SGD (972) but slower than Adam (202). SignMuon reaches the target in 2104 iterations, significantly outperforming SignSGD (3120). EF21-MuonUSign requires 3694 iterations and EF21-MuonSign requires 3564 iterations; its overhead on this benign convex problem is expected, since its purpose is to guarantee convergence on problems where SignMuon does not converge (Theorem 1). The optimization trajectories are shown in AppendixA.9, Figure 6.

5.2 Centralized Learning

In this experiment, we compare the convergence dynamics and the final accuracies of SignMuon, EF21-MuonUSign and EF21-MuonSign with those of the classical optimizers: Muon, SignSGD, SGD, and Adam at a fixed number of epochs. One of the goals of the experiment is to demonstrate that SignMuon achieves classification quality comparable to Muon, while outperforming the classical SignSGD algorithm under the same uplink communication budget (1 bit per parameter on the uplink).

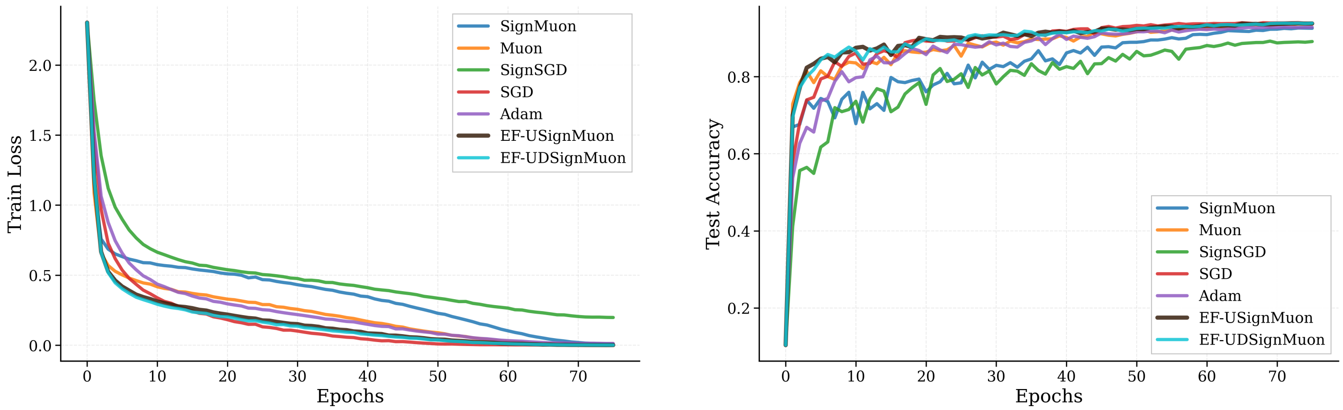
The experiments are conducted on CIFAR-10 (Krizhevsky, Hinton et al. 2009). It is a standard image classification dataset. It contains 60,000 color images of 10 classes. Each image is represented as a tensor of size $3 \times 32 \times 32$, where the first dimension corresponds to the number of color channels (RGB). For centralized training, the dataset is split into training and test sets. In federated learning, the training set is partitioned across the clients.

The base model is a ResNet-18 architecture adapted to low-resolution images by using an initial 3×3 convolution without max-pooling. The network consists of 4 residual blocks with Batch Normalization (BatchNorm), followed by a dropout regularization layer and a linear classifier. As mentioned above, training is run for a fixed number of epochs $T_{\max} = 75$. During training, the learning rate is annealed with a cosine scheduler (CosineAnnealingLR). The learning rate η_t at epoch t is updated as follows:

$$\eta_t = \eta_{\text{min}} + \frac{1}{2}(\eta_{\text{max}} - \eta_{\text{min}})\left(1 + \cos\left(\frac{t}{T_{\text{max}}}\pi\right)\right), \quad (19)$$

where η_{max} is the initial learning rate of the algorithm, $\eta_{\text{min}} = 0$ is the minimum value (in our configuration), and T_{max} is the total number of training epochs.

The experimental results and the selected hyperparameter values are reported in Appendix in Table 6. The dynamics of the loss function and of the accuracy of the algorithms



(a) Train Loss

(b) Test Accuracy

Figure 2: Centralized setting. Comparison of convergence and classification accuracy achieved by different optimization algorithms while training a ResNet-18 classifier on CIFAR-10.

during training are illustrated in Figure 2. Notably, EF21-MuonUSign achieves 93.82% test accuracy, surpassing SignMuon (92.55%) and matching Muon (93.70%), confirming that the Error-Feedback mechanism does not sacrifice practical accuracy.

5.3 Federated Learning

The next stage of the research was adaptation of SignMuon in federated learning, where transmitting full 32-bit gradient matrices is the communication bottleneck. The goal of this stage is to confirm that extreme compression of the orthogonalized updates attains an accuracy comparable to that of Muon, while significantly reducing the volume of transmitted data.

For these experiments, we implemented a federated architecture with a central aggregation server. To keep the results comparable with the previous stages, we used the convolutional neural network (CNN2), comprising two convolutional blocks with BatchNorm and an MLP classifier. The CIFAR-10 dataset was partitioned across the clients homogeneously (homogeneous split). Within this learning type, we study both the baseline Federated SignMuon scheme (Algorithm 8) and the error-feedback variant Federated EF21-MuonUSign (Algorithm 9). Each communication round consists of a local gradient accumulation step on the clients, followed by transmitting the sign-compressed updates to the server, which updates the global model.

For a systematic analysis, we compare the proposed methods against classical baselines at two federation scales. At each scale we evaluate the proposed SignMuon and the Error-Feedback variants EF21-MuonUSign and EF21-MuonSign (Algorithms 8 and 9) against the Muon, SignSGD, SGD, and Adam baselines.

Experiment 1: Comparison of the algorithms with $N = 3$ clients. We compare the optimizers with 3 clients, 2000 rounds, and 1 local step, a regime of limited federation resources. The results are reported in AppendixA.11.

Experiment 2: Comparison of the algorithms with $N = 10$ clients. The potential of sign-based optimization is most fully revealed when the scale of the network and the amount of local computation increase, so at the final stage we increase the number of clients to 10 and use 3 local steps per client. In this regime SignMuon reaches the quality of Muon: with a sufficient number of clients ($N \geq 10$), SignMuon achieves a quality comparable to Muon while reducing the volume of transmitted data by a factor of $\sim 32\times$, by transmitting only the signs of the gradients instead of their full values. The results are reported in Table 2 and Figure 8. EF21-MuonUSign (84.59%) performs on par with SignMuon (84.57%), and the bidirectional EF21-MuonSign (84.87%) matches them as well; at this scale the convergence guarantees cost no accuracy. All methods are evaluated, here and throughout, on the exact server model $\mathbf{X}_t$ – the iterate Corollary 1 bounds – which for EF21-MuonSign is not the model the clients hold. The two coincide on ResNet-18 layer shapes but need not in general, since the downlink contraction degrades with the layer rank (Remark 6); the next section measures that on much wider matrices.

5.4 Language Modelling

We test the methods where the matrices are large enough for the layer-rank dependence of Corollary 1 to bite: the modded-nanoGPT speedrun (record #40), a 12-layer transformer of model dimension 768 with 768×3072 hidden matrices, trained on 611M FineWeb tokens on 8×H100. All eight methods replace the record's Muon on the hidden matrices and gates; embeddings, scalars and the head keep the record's Adam. Full hyperparameters are in Appendix A.13. One η_0 per *family* is used – 0.06 for the LMO-terminated methods, which is the record's own value, and 0.03 for the SIGN-terminated ones – fixed a priori by the unit-gain rule of Appendix A.14 rather than tuned per method; this keeps every contrast a matched-hyperparameter one, at the cost of not reporting each method's best achievable loss. Runs are single-seed, and Table 3 states the seed noise of the reference so that the resolvable differences are explicit.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	10	3	64	0.001	0.9	84.57%
	MuonUSign					-		-%
	Muon					0.05		85.22%
	SignSGD					0.001		78.98%
	SGD					0.03		81.38%
	Adam					0.001		78.36%
	EF21-MuonUSign					0.02		84.59%
	EF21-MuonSign					0.035		84.87%

Table 2: CIFAR-10: federated learning. Experiment 2: Comparison of the algorithms with $N = 10$ clients.

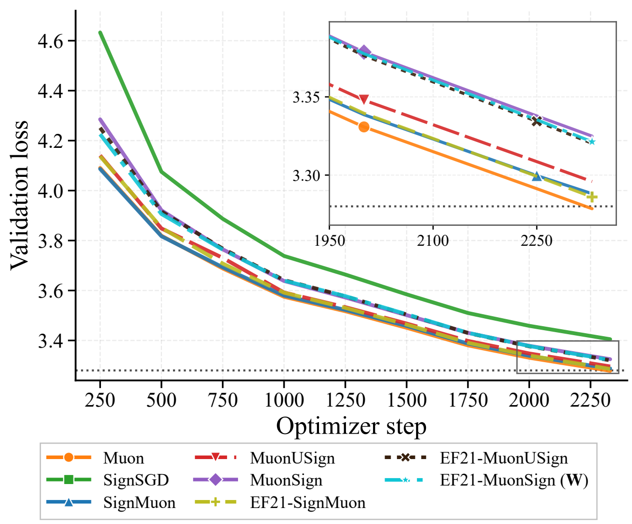


Figure 3: NanoGPT speedrun, 8×H100: validation loss against optimizer step; dotted is the speedrun target 3.28. Our Muon arm *is* record #40's own optimizer and reproduces its published result to 0.6 sd (Table 3), which validates the port before anything else is read. EF21-MuonSign is drawn at its broadcast model $\mathbf{W}$. Marker positions are staggered so that curves lying within a line width of one another can still be followed individually; the inset magnifies the boxed tail, where SignSGD is above the range shown. MuonSign, EF21-MuonUSign and EF21-MuonSign span only 0.005 nats, so Table 3 is the place to read exact values.

Method	**Step**	η_0	**Val. loss**	**Steps to** 3.35
Muon (record #40)	LMO	0.06	**3.2785**	1.00×
EF21-SignMuon	LMO	0.06	3.2860	1.02×
SignMuon	SIGN	0.03	3.2881	1.02×
MuonUSign	LMO	0.06	3.2959	1.05×
EF21-MuonUSign	LMO	0.06	3.3203	1.13×
EF21-MuonSign ($\mathbf{W}$)	LMO	0.06	3.3213	1.14×
MuonSign	SIGN	0.03	3.3249	1.14×
SignSGD	SIGN	0.03	3.4049	–
EF21-MuonSign ($\mathbf{X}$)	LMO	0.06	5.5198	–

Table 3: NanoGPT after 2330 steps (611M tokens). Upstream record #40 reports 3.2780 ± 0.0009 over five seeds, so differences below $\sim$0.003 are noise. "Steps to 3.35" is the step count to reach that loss, relative to Muon, i.e. the compute overhead. Wall-clock is within 1% across all eight (61.4–62.1 ms/step). The last row is EF21-MuonSign's exact server model $\mathbf{X}$; the row above it is the same run's broadcast model $\mathbf{W}$.

Three things follow (Table 3, Figure 3). First, the matrix-aware 1-bit methods transfer to language modelling: all of them beat SignSGD by 0.08–0.12 nats, and the best of them sit within 0.01 of full-precision Muon, at equal wall-clock. Second, error feedback is cheap but not free. Its clean cost is the matched pair MuonUSign $\to$ EF21-MuonUSign, $+0.024$ nats or $\sim 8\%$ more steps – the price of steering the LMO with an estimator that lags its target. EF21-SignMuon, whose target $\mathrm{polar}(\tilde{\mathbf{M}}_t)$ is orthogonal and hence isotropic, pays almost nothing and is the strongest compressed method here, notwithstanding Theorem 4: divergence there is a property of a constructed instance, not of transformer training.

Third, the bidirectional method separates into two models. Its broadcast iterate $\mathbf{W}$ – the one every gradient is evaluated at – is indistinguishable from EF21-MuonUSign (3.3213 vs 3.3203), so the compressed *downlink* costs nothing where training happens. Its exact server iterate $\mathbf{X}$, which is what Corollary 1 bounds, reaches only 5.52 and rises monotonically after step 750. The diagnostic (Figure 10) localizes this to one layer type: on the zero-initialized `c_proj` the scaled sign achieves $\alpha(\boldsymbol{\Delta}^{\downarrow}) = 1.2 \times 10^{-4}$ instead of the $\Theta(10^{-1})$ it achieves everywhere else, and the gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F / \|\mathbf{W}_t\|_F$ stalls at 8% against $<1\%$ for every other layer. This is the mechanism of Remark 6 rather than a tuning failure: a layer built from zero receives maximally correlated updates, its downlink residual concentrates, and a compressor that moves every coordinate by $\mathrm{mean}\,|\boldsymbol{\Delta}^{\downarrow}|$ cannot catch the coordinates that are driven hardest. Lowering η_0 does not repair it, because the admissible step size would have to shrink by the layer rank (Remark 7); a downlink compressor contractive in the *spectral* norm would.

6 Conclusion

This work proposes SignMuon, which applies 1-bit sign compression to the Muon LMO direction to bridge structured optimization and extreme communication efficiency in federated learning. While it empirically matches Muon's accuracy and outperforms SignSGD, we formally prove that direct sign-compressed LMO lacks global convergence guarantees and can diverge. To resolve this without compromising uplink efficiency, we introduce EF21-MuonUSign, an Error-Feedback variant that applies sign compression to the internal residual rather than the LMO step. EF21-MuonUSign restores convergence on our counterexample and achieves Muon-level performance while maintaining a 1-bit uplink. Our framework establishes a foundation for adapting matrix-aware optimizers to bandwidth-constrained distributed training without sacrificing model accuracy.

Future work. Our (L^0, L^1)-smoothness guarantee (Corollary 2) covers the uplink-compressed EF21-MuonUSign only, since the generalized-smooth theory of Gruntkowska et al. (2025) is currently available only without downlink compression; extending it to the bidirectional EF21-MuonSign is an open theoretical question. A second question concerns the downlink. The rank penalty EF21-MuonSign pays comes from compressing the model in the wrong norm (Remark 7) and is removable only by changing the compressor: random dropout ($\alpha_P = p$) and the Top-K SVD compressor ($\alpha_P = 1 - \sigma_{K+1}^2/\sigma_1^2$) are both admissible at a comparable budget, but neither is strictly one-bit. Whether *any* one-bit compressor can be layer-norm contractive we leave open. Since Gruntkowska et al. (2025) fix the server compressor to the identity throughout their experiments, this regime appears not to have been probed empirically before. On the practical side, the reduction of Appendix A.7 applies verbatim to arbitrary layer geometries (EF21-GluonUSign and EF21-GluonSign, Remark 8), and investigating how these matrix-aware 1-bit strategies scale to the massive parameter matrices of Large Language Models is a highly promising direction for future research.

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A Appendix

A.1 Divergence on linear objectives: reduction and momentum

Proof of Proposition 1. On the linear objective (10) the gradient is globally constant, $\mathbf{G}_t = \mathbf{G}$, so the momentum buffer of (7) is $\mathbf{M}_t = \sum_{i=0}^{t-1} \mu^i \, \mathbf{G} = \frac{1-\mu^t}{1-\mu} \, \mathbf{G}$. Both momentum rules then return a positive multiple of $\mathbf{G}$,

$$\tilde{\mathbf{M}}_t = \gamma_t \, \mathbf{G}, \qquad \gamma_t = \begin{cases} \frac{1-\mu^t}{1-\mu}, & \text{(Standard)}, \\ 1 + \mu \, \frac{1-\mu^t}{1-\mu}, & \text{(Nesterov)}, \end{cases} \tag{20}$$

because $\mu \in [0, 1)$. The elementwise $\operatorname{sign}(\cdot)$ and the Muon LMO $\operatorname{polar}(\cdot)$ are each invariant under multiplication by a positive scalar, so evaluating (9) at $\tilde{\mathbf{M}}_t = \gamma_t \mathbf{G}$ returns the same matrix as evaluating it at $\mathbf{G}$; that is, $\mathbf{s}_t = \mathbf{s}(\mathbf{G})$ for every t, independently of μ and of the momentum variant. Hence $f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = \langle \mathbf{G}, \mathbf{X}_{t+1} - \mathbf{X}_t \rangle = -\eta_t \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle$, which is (11). If $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$, then $f(\mathbf{X}_t) = f(\mathbf{X}_0) - \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle \sum_{i=0}^{t-1} \eta_i$ increases strictly at every step and diverges to $+\infty$ whenever $\sum_t \eta_t = \infty$. ■

By Proposition 1, each divergence theorem reduces to exhibiting a single gradient $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$; the three proofs below do exactly this, and momentum needs no further comment.

A.2 Proof of Theorem 1 (Divergence of SignMuon)

For SignMuon $\mathbf{s}(\mathbf{G}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G}))$, so by (11) it suffices to construct $\mathbf{G} \in \mathbb{R}^{4 \times 4}$ with $\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \rangle < 0$. An exhaustive numerical search found no such $\mathbf{G}$ for $N_{dim} \leq 3$; the guarantee first fails at $N_{dim} = 4$.

We construct $\mathbf{G} \in \mathbb{R}^{4 \times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1 \mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103} \begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \tag{21}$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \operatorname{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}} \begin{pmatrix} 10 \\ -3 \\ 10 \\ 10 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}} \begin{pmatrix} 10 \\ 3 \\ -10 \\ 10 \end{pmatrix}. \tag{22}$$

One can easily verify that $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, meaning $\mathbf{u}_1$ and $\mathbf{v}_1$ act perfectly as left and right singular vectors for this orthogonal space. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1 \mathbf{v}_1^\top, \mathbf{S} \rangle = \mathbf{u}_1^\top \mathbf{S} \mathbf{v}_1 = -\frac{43}{103}. \tag{23}$$

We construct the gradient matrix $\mathbf{G}$ by assigning a massive singular value ($\sigma_1 = 1001$) to this pathological component and a singular value of 1 to the rest of the orthogonal space:

$$\mathbf{G} \approx \begin{pmatrix} 324.6 & 97.3 & -323.6 & 323.6 \\ -97.3 & -28.2 & 97.3 & -97.3 \\ 323.6 & 97.3 & -323.6 & 324.6 \\ 323.6 & 97.3 & -324.6 & 323.6 \end{pmatrix}. \tag{24}$$

Indeed, since $\mathbf{G} = 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}$ and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$,

$$\begin{aligned} \mathbf{G}\mathbf{v}_1 &= 1000\,\mathbf{u}_1(\mathbf{v}_1^\top\mathbf{v}_1) + \mathbf{O}\mathbf{v}_1 \\ &= 1001\,\mathbf{u}_1, \end{aligned}$$

so $(\mathbf{u}_1, \mathbf{v}_1)$ is a singular pair of $\mathbf{G}$ with $\sigma_1 = 1001$. For any $\mathbf{w} \perp \mathbf{v}_1$ we have $\mathbf{G}\mathbf{w} = \mathbf{O}\mathbf{w}$; as $\mathbf{O}$ is orthogonal and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, the restriction $\mathbf{O}|_{\mathbf{v}_1^\perp}$ is an isometry onto $\mathbf{u}_1^\perp$, hence $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Moreover, with $\mathbf{u}_k = \mathbf{O}\mathbf{v}_k$ for $k \geq 2$,

$$\textstyle\sum_{k\geq 2} \mathbf{u}_k\mathbf{v}_k^\top = \mathbf{O}\big(\mathbf{I} - \mathbf{v}_1\mathbf{v}_1^\top\big) = \mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top,$$

so $\mathbf{U}_G\mathbf{V}_G^\top = \mathbf{u}_1\mathbf{v}_1^\top + (\mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top) = \mathbf{O}$.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

$$\langle \mathbf{G}, \mathbf{S}\rangle = \langle 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}, \mathbf{S}\rangle = 1000\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle + \langle \mathbf{O}, \mathbf{S}\rangle. \tag{25}$$

The inner product $\langle \mathbf{O}, \mathbf{S}\rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S}\rangle = 1000\left(-\frac{43}{103}\right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \tag{26}$$

Here $\mathbf{S} = \mathrm{sign}(\mathbf{O}) = \mathrm{sign}(\mathrm{polar}(\mathbf{G})) = \mathbf{s}(\mathbf{G})$, so $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle = -\frac{42468}{103} < 0$. By Proposition 1, SignMuon strictly ascends, $f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = \frac{42468}{103}\,\eta_t > 0$ at every iteration, for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.3 Proof of Theorem 2 (Divergence of MuonUSign)

MuonUSign (Algorithm 4) applies the sign *before* the LMO, so $\mathbf{s}(\mathbf{G}) = \mathrm{polar}(\mathrm{sign}(\mathbf{G}))$. By (11) it suffices to construct $\mathbf{G} \in \mathbb{R}^{5\times 5}$ with $\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\rangle < 0$.

Fix the full-rank sign matrix $\mathbf{S} \in \{-1, 1\}^{5\times 5}$,

$$\mathbf{S} = \begin{pmatrix} -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 & 1 \end{pmatrix}, \tag{27}$$

and let $\mathbf{D} = \mathrm{polar}(\mathbf{S})$ be its (unique, since $\mathbf{S}$ is full rank) Muon LMO direction. A direct computation gives $D_{4,2} = -0.2425 < 0$ while $\mathrm{sign}(D_{i,j}) = S_{i,j}$ at all 24 other entries; equivalently, $\mathrm{sign}(\mathbf{D})$ and $\mathbf{S}$ disagree at exactly the single entry $(4, 2)$, where $S_{4,2} = +1$. We exploit this lone mismatch. Define

$$\mathbf{G} = \epsilon\,\mathbf{S} + (M - \epsilon)\,\mathbf{e}_4\mathbf{e}_2^\top, \qquad \epsilon > 0,\ M > \epsilon, \tag{28}$$

so that $G_{4,2} = M > 0$ and $G_{i,j} = \epsilon S_{i,j}$ otherwise; hence $\mathrm{sign}(\mathbf{G}) = \mathbf{S}$ and $\mathrm{polar}(\mathrm{sign}(\mathbf{G})) = \mathbf{D}$ for every $M > 0$. The descent inner product splits as

$$\langle \mathbf{G}, \mathbf{D}\rangle = \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j}D_{i,j} \;+\; M\,D_{4,2} = \epsilon\,C + M\,D_{4,2}, \tag{29}$$

where $C := \sum_{(i,j)\neq(4,2)} |D_{i,j}| = 10.366 > 0$ is fixed (every such entry agrees in sign with $\mathbf{S}$), while $M\,D_{4,2} = -0.2425\,M \to -\infty$. Thus $\langle \mathbf{G}, \mathbf{D}\rangle < 0$ for any $M > C\epsilon/0.2425$; with $\epsilon = 1,\ M = 100$ one obtains $\langle \mathbf{G}, \mathbf{D}\rangle = -13.89$, which we verified for both the exact polar factor and the Newton–Schulz approximation used in practice. By Proposition 1, MuonUSign strictly ascends on $f(\mathbf{W}) = \langle \mathbf{G}, \mathbf{W}\rangle$ for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.4 Proof of Theorem 3 (Divergence of MuonSign)

MuonSign (Algorithm 5) signs the LMO output as well, so $\mathbf{s}(\mathbf{G}) = \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G}))) = \mathrm{sign}(\mathbf{D})$. We reuse the *same* $\mathbf{S}$ and $\mathbf{G}$ of (27)–(28): since $\mathrm{sign}(\mathbf{G}) = \mathbf{S}$, the bidirectional step is the constant matrix $\mathrm{sign}(\mathbf{D})$, which agrees with $\mathbf{S}$ at all 24 entries except $(4, 2)$, where $\mathrm{sign}(D_{4,2}) = -1 = -S_{4,2}$. Using $S_{i,j}^2 = 1$ everywhere,

$$\begin{aligned} \langle \mathbf{G}, \mathrm{sign}(\mathbf{D})\rangle &= \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j}\,\mathrm{sign}(D_{i,j}) + M\ \mathrm{sign}(D_{4,2}) \\ &= 24\,\epsilon - M. \end{aligned} \tag{30}$$

This is negative for every $M > 24\epsilon$; with $\epsilon = 1,\ M = 100$ it equals exactly -76. By Proposition 1, MuonSign strictly ascends on $f(\mathbf{W}) = \langle \mathbf{G}, \mathbf{W}\rangle$ for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. In particular, the *same* 5×5 linear instance defeats both the uplink-only (MuonUSign) and the bidirectional (MuonSign) placements. ■

A.5 Comparison with the convergence claim for SignMuon

Concurrently, Mishra, Trivedi, and Kumar (2026) introduced the same sign-after-LMO method under the name *SignMuon*, with update direction $\mathbf{S}_t = \mathrm{sign}(\mathrm{polar}(\mathbf{M}_t))$ rescaled to $\mathbf{D}_t = \mathbf{S}_t/\sqrt{mn}$; this coincides with the SignMuon step of Section 4 after absorbing $1/\sqrt{mn}$ into η. For this method they establish an $\mathcal{O}(1/\sqrt{T})$ stationarity guarantee. We show that this guarantee does not contradict Theorem 1: the finalized rate is proved for a different update direction, and for $\mathbf{D}_t$ their bound is uninformative on the instance of Theorem 1.

Their analysis controls the stationarity proxy $\mathcal{G}_T = \frac{1}{T}\sum_t \mathbb{E}[\|\mathbf{g}_t\|_1/\sqrt{mn}]$, where $\mathbf{g}_t = \nabla f(\mathbf{W}_t)$, through the per-entry sign-error probabilities $q_{ij,t} = \Pr(\mathbf{S}_{t,ij} \neq \mathrm{sign}(\mathbf{g}_{t,ij}) \mid \mathbf{W}_t)$, in two steps. First, for an arbitrary sign oracle $\mathbf{S}_t$ they derive

$$\mathcal{G}_T \le \frac{f(\mathbf{W}_0)-f^*}{\eta T} + \frac{L_*}{2}\,\eta + \underbrace{\frac{2}{T}\sum_t \mathbb{E}\left[\frac{1}{\sqrt{mn}}\sum_{i,j}|\mathbf{g}_{t,ij}|\,q_{ij,t}\right]}_{=:\,R_T}. \tag{31}$$

Inequality (31) applies in particular to $\mathbf{S}_t = \mathrm{sign}(\mathrm{polar}(\mathbf{M}_t))$, but with the residual R_T left unbounded. Second, the rate is obtained by estimating R_T, and this estimate is derived *only* for the gradient-sign oracle $\mathbf{S}_t = \mathrm{sign}(\widetilde{\mathbf{G}}_t)$, where $\widetilde{\mathbf{G}}_t$ is an unbiased stochastic gradient with coordinatewise variance $\mathrm{Var}(\widetilde{\mathbf{G}}_{t,ij} \mid \mathbf{W}_t) \le \sigma_{ij}^2/n_b$ and n_b the mini-batch size. In that case a Markov–Jensen argument yields $|\mathbf{g}_{t,ij}|\,q_{ij,t} \le \sigma_{ij}/\sqrt{n_b}$, hence $R_T = \mathcal{O}(\|\sigma\|_1/\sqrt{mn\,n_b})$ with $\|\sigma\|_1 = \sum_{i,j}\sigma_{ij}$, which vanishes as $n_b \to \infty$.

This estimate bounds $q_{ij,t}$ by the noise level $\sigma_{ij}/\sqrt{n_b}$ and is therefore specific to $\mathrm{sign}(\widetilde{\mathbf{G}}_t)$, whose only source of sign error is gradient noise. It does not extend to the polar-sign oracle: even with exact gradients ($\sigma_{ij}=0$) the latter satisfies $q_{ij,t} = \mathbf{1}[\mathrm{sign}(\mathrm{polar}(\mathbf{g}_t))_{ij} \ne \mathrm{sign}(\mathbf{g}_{t,ij})]$, which need not vanish. Consequently the finalized $\mathcal{O}(1/\sqrt{T})$ guarantee holds for the gradient-sign update $\mathbf{S}_t = \mathrm{sign}(\widetilde{\mathbf{G}}_t)$, whose direction never invokes the polar factor, and not for the LMO-based direction $\mathrm{sign}(\mathrm{polar}(\mathbf{M}_t))$ to which it is attributed.

On the instance of Theorem 1 the residual R_T is not merely unbounded: it exceeds $\mathcal{G}_T$. Consider the linear objective (10) with the 4×4 gradient $\mathbf{G}$ of Theorem 1. By Proposition 1 one has $\mathbf{g}_t \equiv \mathbf{G}$ and $\mathrm{polar}(\mathbf{M}_t) = \mathrm{polar}(\mathbf{G})$ at every iteration, so the oracle is deterministic and the inner-product identity $\mathbb{E}[\langle \mathbf{g}_t, \mathbf{D}_t\rangle \mid \mathbf{W}_t] = \frac{1}{\sqrt{mn}}\sum_{ij}|\mathbf{g}_{t,ij}|(1-2q_{ij,t})$ underlying (31) reduces to

$$\sum_{i,j}|\mathbf{G}_{ij}|\,(1-2q_{ij}) = \big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G}))\big\rangle = -\tfrac{42468}{103} < 0. \tag{32}$$

Therefore $\sum_{ij}|\mathbf{G}_{ij}|\,q_{ij} > \frac{1}{2}\|\mathbf{G}\|_1$, and at every iteration

$$\frac{2}{\sqrt{mn}}\sum_{i,j}|\mathbf{G}_{ij}|\,q_{ij} \;>\; \frac{\|\mathbf{G}\|_1}{\sqrt{mn}} \;=\; \mathcal{G}_t. \tag{33}$$

The unestimated term R_T in (31) thus strictly exceeds the left-hand side, so the bound reduces to $\mathcal{G}_T \le (\text{nonnegative}) + R_T$ with $R_T > \mathcal{G}_T$ and imposes no constraint on $\mathcal{G}_T$. In agreement, (32) yields the exact per-step increase $f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = -\eta\langle \mathbf{G}, \mathbf{D}_t\rangle = \frac{\eta}{4}\cdot\frac{42468}{103} > 0$ for every $\eta > 0$, which is the divergence established in Theorem 1.

A.6 Proof of Theorem 4 (Divergence of EF21-SignMuon)

Recall from the main text that EF21-SignMuon (Algorithm 3) does not descend along the LMO output $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$ itself, but along the error-feedback estimate $\mathbf{d}_t^{\mathrm{est}}$ of (12), followed by $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta\,\mathbf{d}_t^{\mathrm{est}}$. A *single* magnitude α_t rescales the signs of *all* entries at once; this coupling is what the counterexample exploits. We now prove Theorem 4.

Proof idea Picture the target $\mathbf{D}_t$ as a 2×2 matrix whose off-diagonal is large and *flips sign every step*, while its diagonal is a small constant. The estimator (12) rescales all four signs by the same α_t, so the flipping off-diagonal pins α_t at $\Theta(1)$, which exceeds the small diagonal target. The diagonal of $\mathbf{d}_t^{\mathrm{est}}$ therefore never settles: it oscillates with $\Theta(1)$ amplitude, and its time-average has the sign *opposite* to the target. Being an average over an oscillation, this bias depends only on the oscillation's phase, which one-bit sign feedback cannot correct. It pushes one entry of $\mathbf{X}_t$ monotonically to infinity (Figure 4).

Two observations make the result independent of the parameters. First, the step size and smoothness only rescale the orbit, so we may fix $\eta = 1$ and one value of L. Second, the divergence lives entirely in the estimator's response to a fixed target sequence; momentum affects only *how the gradients are sourced*, not the targets. A single explicit trajectory thus settles all $(L, \eta, \mu, \text{variant})$ at once. The mechanism is not degeneracy: the matrices driving it have condition number $\frac{16}{9}$ throughout.

Accordingly the proof has three parts. Part 1 rescales the problem to remove L and η. Part 2 isolates the dynamical core: for one fixed sequence of LMO targets, the estimator provably converges to a limit cycle whose diagonal has the wrong sign. Part 3 exhibits a concrete smooth function whose gradients generate that target sequence, for every μ and either momentum variant.

Proof

Part 1: rescaling.

Lemma 1 (Scale reduction) *If $\tilde{f}$ is $\tilde{L}$-smooth with EF21-SignMuon iterates $\tilde{\mathbf{X}}_t$ at step size 1, then $f(\mathbf{X}) := \frac{L\eta^2}{\tilde{L}}\,\tilde{f}(\mathbf{X}/\eta)$ is L-smooth and its run at step size η (same μ, same variant, from $\mathbf{0}$) satisfies $\mathbf{X}_t = \eta\,\tilde{\mathbf{X}}_t$ and $f(\mathbf{X}_t) = \frac{L\eta^2}{\tilde{L}}\,\tilde{f}(\tilde{\mathbf{X}}_t)$.*

Proof. $\nabla f(\mathbf{X}) = \frac{L\eta}{\tilde{L}}\nabla\tilde{f}(\mathbf{X}/\eta)$, so f is L-smooth. Suppose $\mathbf{X}_s = \eta\tilde{\mathbf{X}}_s$ for $s < t$: the two gradients then differ by the positive factor $\frac{L\eta}{\tilde{L}}$, which $\mathbf{M}_t$ and $\tilde{\mathbf{M}}_t$ (positive combinations of past gradients) inherit and polar discards. Hence $\mathbf{D}_t$ and $\mathbf{d}_t^{\mathrm{est}}$ match the normalized run, and $\mathbf{X}_t = \eta\tilde{\mathbf{X}}_t$. ■ So it suffices to exhibit, for each $(\mu, \text{variant})$, a C^∞ $\tilde{L}$-smooth $\tilde{f}$ on which the *normalized* run ($\eta = 1$) obeys (13); that $\tilde{f}$ may be taken bounded below (Assumption 1) is shown afterwards (Remark 1). We fix $\eta = 1$ from now on.

Part 2: the estimator's limit cycle. The dynamical core of the proof is the behavior of the estimator (12) on the fixed target sequence

$$\mathbf{D}_1 = \mathbf{S}_1, \quad \mathbf{D}_2 = \mathbf{S}_2, \quad \mathbf{D}_t = \bar{\mathbf{D}}^{(-1)^t}\ (t \ge 3), \tag{34}$$

$$\bar{\mathbf{D}}^{\pm} = \begin{pmatrix} 7/25 & \pm 24/25 \\ \pm 24/25 & -7/25 \end{pmatrix}, \qquad \mathbf{S}_1 = \begin{pmatrix} -4/5 & 3/5 \\ -3/5 & -4/5 \end{pmatrix}, \quad \mathbf{S}_2 = \begin{pmatrix} 3/5 & -4/5 \\ 0 & 0 \end{pmatrix}. \tag{35}$$

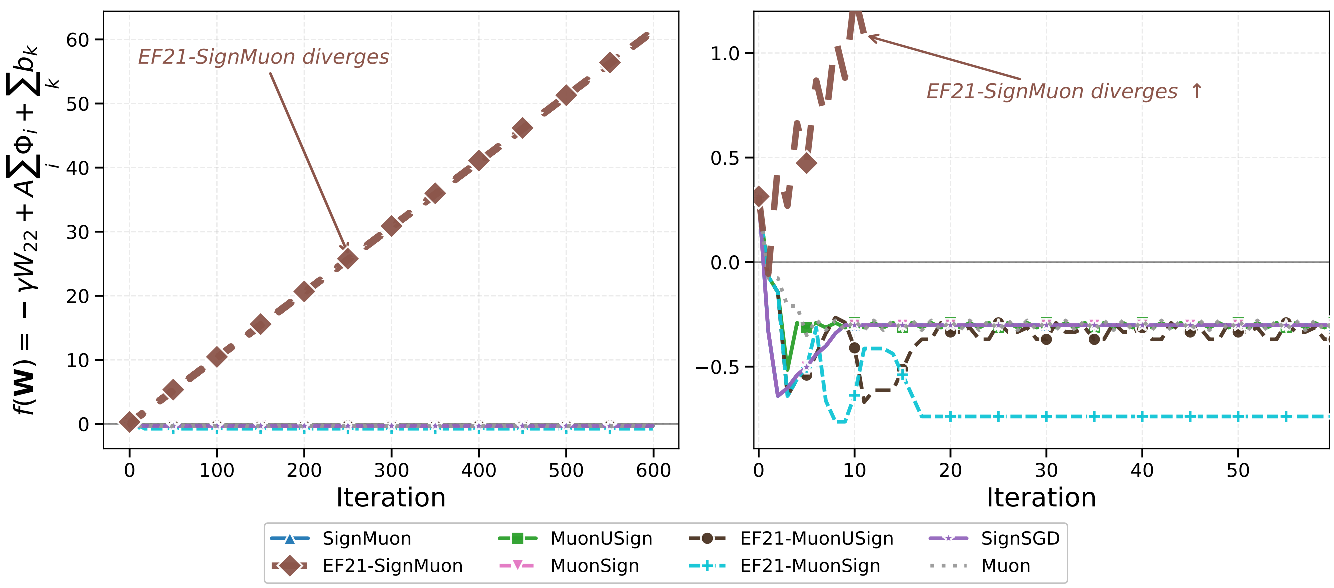


Figure 4: All eight methods on the 2×2 instance of Theorem 4 (normalized units, $\eta = 1$, $\mu = 0$). *Left:* full scale — EF21-SignMuon is the only method whose objective increases, rising at the exact rate $\frac{49}{480}$ per step. *Right:* the same runs magnified to the band of the seven remaining methods (EF21-SignMuon leaves the frame): SignMuon, MuonUSign, MuonSign, EF21-MuonUSign, EF21-MuonSign, SignSGD and Muon all decrease and stay bounded. By Lemma 1 the EF21-SignMuon trajectory is the same for every $\mu \in [0, 1)$ and both momentum variants, so the single run shown is representative of all of them.

The divergence originates in the tail ($t \geq 3$): the reflections $\bar{\mathbf{D}}^{\pm}$ share the diagonal $(\frac{7}{25}, -\frac{7}{25})$, while their off-diagonal $\pm\frac{24}{25}$ reverses sign at every step. The rotation $\mathbf{S}_1$ and the rank-one $\mathbf{S}_2$ form a two-step preamble whose only role is to place the estimator in the divergent phase of the cycle.

Lemma 2 (Wrong-sign limit cycle) *On the targets* (34)*, the recursion* (12) *from* $\mathbf{d}_0^{\text{est}} = \mathbf{0}$ *enters at* $t = 3$ *the exact period-two cycle* $\mathbf{d}_t^{\text{est}} = \mathbf{d}_B$ *(odd* t*),* $\mathbf{d}_A$ *(even* t*), where*

$$\mathbf{d}_B = \tfrac{1}{200}\begin{pmatrix} -61 & -201 \\ -61 & -61 \end{pmatrix}, \qquad \mathbf{d}_A = \tfrac{1}{200}\begin{pmatrix} 131 & -9 \\ 131 & 131 \end{pmatrix}. \tag{36}$$

Its $(2,2)$*-entry averages* $\frac{1}{2}\big((\mathbf{d}_A)_{22} + (\mathbf{d}_B)_{22}\big) = +\frac{7}{40}$*, opposite in sign to every target value* $(\mathbf{D}_t)_{22} = -\frac{7}{25}$*.*

Proof. Substituting (34) into (12) gives the values

t	1	2	3	4 (then 2-periodic)
α_t	$\frac{7}{10}$	$\frac{21}{20}$	$\frac{131}{200}$	$\frac{24}{25}$
$\mathbf{d}_t^{\text{est}}$	$\frac{1}{10}\begin{pmatrix} -7 & 7 \\ -7 & -7 \end{pmatrix}$	$\frac{1}{20}\begin{pmatrix} 7 & -7 \\ 7 & 7 \end{pmatrix}$	$\mathbf{d}_B$	$\mathbf{d}_A$

(37)

Each residual $\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}$ has strictly nonzero entries, so the signs are unambiguous; e.g. at $t = 3$ it is $\frac{1}{100}\begin{pmatrix} -7 & -61 \\ -131 & -63 \end{pmatrix}$, all negative. From $t = 3$ the targets are 2-periodic and the pair $(\mathbf{d}_B, \mathbf{d}_A)$ reproduces itself: $\bar{\mathbf{D}}^{+} - \mathbf{d}_B$ and $-(\bar{\mathbf{D}}^{-} - \mathbf{d}_A)$ are both entrywise positive with mean $\frac{24}{25}$, so $\mathbf{d}_B + \frac{24}{25}\mathbf{J} = \mathbf{d}_A$ and $\mathbf{d}_A - \frac{24}{25}\mathbf{J} = \mathbf{d}_B$ ($\mathbf{J}$ all-ones). ■

Since $\mathbf{X}_t = \mathbf{X}_{t-1} - \mathbf{d}_t^{\text{est}}$ (recall $\eta = 1$), over one period the $(2,2)$-coordinate changes by $-(\mathbf{d}_A + \mathbf{d}_B)_{22} = -\frac{7}{20}$. Consequently a term $-\gamma W_{22}$ in $\tilde{f}$, with $\gamma := \frac{7}{12}$, increases by $\gamma \cdot \frac{7}{20} = \frac{49}{240}$ per period. This is the divergence, provided a genuine smooth function produces the targets (34); Part 3 constructs one.

Part 3: realization by a smooth function. Fix $(\mu, \text{variant})$ and set $\gamma = \frac{7}{12}$, $\nu = \frac{1}{1+2\mu}$, and

$$A = \frac{1+\mu}{1-\mu} \ \text{(standard)}, \qquad A = \frac{1+\mu}{(1-\mu)(1+2\mu)} \ \text{(Nesterov)}. \tag{38}$$

We build $\tilde{f} = g + \sum_{k=1}^{3} b_k$ from a periodic-plus-linear *field* g and three *localized corrections* b_k, all explicit.

The field is

$$g(\mathbf{W}) = -\gamma\, W_{22} + A\,\Phi_1(W_{12}) + A\,\Phi_2(W_{21}), \tag{39}$$

where $\Phi_i(w) = \int_0^w \psi_i$ and $\psi_i : \mathbb{R} \to [-1, 1]$ is a fixed C^∞, p_i-periodic function with $\int_0^{p_i} \psi_i = 0$,

$$p_1 = \tfrac{21}{20}, \quad p_2 = \tfrac{7}{20}, \qquad \delta = \tfrac{1}{100},$$

equal to $+1$ on $[\rho_i^+ - \delta, \rho_i^+ + \delta]$ and -1 on $[\rho_i^- - \delta, \rho_i^- + \delta]$ (mod p_i), where

$$\rho_1^+ = \tfrac{131}{200}, \ \rho_1^- = \tfrac{140}{200}; \qquad \rho_2^+ = \tfrac{61}{200}, \ \rho_2^- = 0.$$

The two bands are disjoint mod p_i (their centers are $\frac{9}{200} > 2\delta$ apart), so such a ψ_i exists; the zero-mean condition, met by balancing the rest of the period, makes Φ_i periodic (hence bounded). On each band $\Phi_i' = \psi_i = \pm 1$ exactly.

The corrections pin the first three gradients. Fix once a C^∞ cutoff $\phi : \mathbb{R} \to [0, 1]$ with $\phi \equiv 1$ on $[0, \frac{1}{2}]$ and $\phi \equiv 0$ on $[1, \infty)$, put $r = \frac{1}{50}$, and set

$$b_k(\mathbf{W}) = \big\langle \mathbf{C}_k,\ \mathbf{W} - \mathbf{Z}_k \big\rangle\, \phi\big(\|\mathbf{W} - \mathbf{Z}_k\|_F / r\big), \tag{40}$$

centered at the first three iterates (from Lemma 2)

$$\mathbf{Z}_1 = \mathbf{0}, \quad \mathbf{Z}_2 = \tfrac{1}{10}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right), \quad \mathbf{Z}_3 = \tfrac{1}{20}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right).$$

Each b_k is C^∞, supported in $\{\|\mathbf{W} - \mathbf{Z}_k\|_F \le r\}$; since its linear factor vanishes at $\mathbf{Z}_k$ while $\phi(0) = 1$, $\nabla b_k(\mathbf{Z}_k) = \mathbf{C}_k$. We choose $\mathbf{C}_k := \hat{\mathbf{G}}_k - \nabla g(\mathbf{Z}_k)$ with the explicit $\hat{\mathbf{G}}_k$ of (41) below, so that $\nabla \tilde{f}(\mathbf{Z}_k) = \hat{\mathbf{G}}_k$. Every term of $\tilde{f}$ has a bounded Hessian, so $\tilde{f}$ is C^∞ and $\tilde{L}$-smooth for a finite $\tilde{L}(\mu, \text{variant})$.

Lemma 3 (Realization) *For every $\mu \in [0, 1)$ and either variant, EF21-SignMuon on this $\tilde{f}$ ($\eta = 1$, from $\mathbf{0}$) generates exactly the targets* (34).

Proof. *The required gradients.* Momentum is a causal, positive-weight linear filter of the gradients, and we invert it. Define the *transient gradients*

$$\hat{\mathbf{G}}_t = \frac{\mathbf{M}_t - \mu \mathbf{M}_{t-1}}{1 - \mu} \quad (t = 1, 2, 3;\ \mathbf{M}_0 = \mathbf{0}) \qquad (41)$$

and the *field gradient* $\mathbf{G}^\pm = \left(\begin{smallmatrix} 0 & \pm A \\ \pm A & -\gamma \end{smallmatrix}\right)$, where the prescribed buffer values $\mathbf{M}_{1,2,3}$ are given below. With $\bar{\mathbf{M}}^\pm = \left(\begin{smallmatrix} 0 & \pm 1 \\ \pm 1 & -\gamma \end{smallmatrix}\right)$, the factorization

$$\bar{\mathbf{M}}^\pm = \bar{\mathbf{D}}^\pm \left(\begin{smallmatrix} 24/25 & \mp 7/25 \\ \mp 7/25 & 337/300 \end{smallmatrix}\right) \quad \text{(second factor } \succ 0,\ \det = 1) \qquad (42)$$

shows $\operatorname{polar}(\bar{\mathbf{M}}^\pm) = \bar{\mathbf{D}}^\pm$, with singular values $\frac{3}{4}, \frac{4}{3}$ (the condition number $\frac{16}{9}$ of the idea).

Standard momentum ($\tilde{\mathbf{M}}_t = \mathbf{M}_t$). Take $\mathbf{M}_1 = \mathbf{S}_1$, $\mathbf{M}_2 = \mathbf{S}_2$, $\mathbf{M}_3 = \bar{\mathbf{M}}^-$; then $\hat{\mathbf{G}}_{1,2,3}$ are the explicit matrices (41). Feeding $\mathbf{G}^{(-1)^t}$ for $t \ge 4$ keeps $\mathbf{M}_t = \bar{\mathbf{M}}^{(-1)^t}$, because $(1-\mu)\mathbf{G}^\pm = \bar{\mathbf{M}}^\pm - \mu \bar{\mathbf{M}}^\mp$ with A as in (38). As $\operatorname{polar}(\mathbf{S}_1) = \mathbf{S}_1$ (orthogonal) and $\operatorname{polar}(\mathbf{S}_2) = \mathbf{S}_2$ (rank one), (42) yields the targets (34).

Nesterov momentum ($\tilde{\mathbf{M}}_t = (1 + \mu)\mathbf{M}_t - \mu \mathbf{M}_{t-1}$). Take $\mathbf{M}_1 = \frac{1}{1+\mu}\mathbf{S}_1\mathbf{H}_1$, $\mathbf{M}_2 = \frac{1}{1+\mu}(\mathbf{S}_2 + \mu \mathbf{M}_1)$, $\mathbf{M}_3 = \bar{\mathbf{M}}'^-$ and $\mathbf{M}_t = \bar{\mathbf{M}}'^{(-1)^t}$ ($t \ge 4$), where $\bar{\mathbf{M}}'^\pm = \left(\begin{smallmatrix} 0 & \pm\nu \\ \pm\nu & -\gamma \end{smallmatrix}\right)$, $\mathbf{H}_1 = \operatorname{diag}(1, 1 + T)$, and

$$T = \left(\tfrac{140(1+\mu)^2}{1+2\mu} - 44\right)\tfrac{1+\mu}{117\mu} > 0.$$

Then $\tilde{\mathbf{M}}_1 = \mathbf{S}_1\mathbf{H}_1$ and $\tilde{\mathbf{M}}_2 = \mathbf{S}_2$ have polar factors $\mathbf{S}_1, \mathbf{S}_2$, and $\tilde{\mathbf{M}}_t = \bar{\mathbf{M}}^{(-1)^t}$ for $t \ge 4$ (since $\nu(1 + 2\mu) = 1$). The one nontrivial step is $t = 3$: with $\mathbf{H}_3 := \bar{\mathbf{D}}^- \tilde{\mathbf{M}}_3$ (so $\tilde{\mathbf{M}}_3 = \bar{\mathbf{D}}^- \mathbf{H}_3$, the reflection being an involution), the stated T is exactly the value making $\mathbf{H}_3$ symmetric, and $\mathbf{H}_3 \succ 0$ for all $\mu \in (0, 1)$: under $\mu = \frac{s}{1+s}$ ($s > 0$) the numerators of its two leading minors are polynomials in s with nonnegative coefficients. Hence $\operatorname{polar}(\tilde{\mathbf{M}}_3) = \bar{\mathbf{D}}^-$, completing (34).

The function delivers these gradients. It remains to verify $\nabla \tilde{f}(\tilde{\mathbf{X}}_{t-1}) = \hat{\mathbf{G}}_t$ for $t \le 3$ and $= \mathbf{G}^{(-1)^t}$ for $t \ge 4$. By Lemma 2 the iterates $\tilde{\mathbf{X}}_0, \tilde{\mathbf{X}}_1, \tilde{\mathbf{X}}_2$ are exactly the centers $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3$, where $\nabla \tilde{f} = \hat{\mathbf{G}}_{1,2,3}$ by construction. The three balls are disjoint ($\|\mathbf{Z}_j - \mathbf{Z}_k\|_F \ge \frac{7}{10} > 2r$), and every later iterate has $(1, 2)$-entry $\ge \frac{131}{200}$ while the centers have it ≤ 0, so no ball is re-entered. For $t \ge 4$ the query lies in the field, where

$$\nabla g(\mathbf{W}) = \begin{pmatrix} 0 & A\psi_1(W_{12}) \\ A\psi_2(W_{21}) & -\gamma \end{pmatrix}.$$

The period shift $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t = -(\mathbf{d}_A + \mathbf{d}_B)$ advances W_{12} by $+p_1$ and W_{21} by $-p_2$ (one full period each), so the off-diagonal coordinates keep the residues ρ_i^+ at odd indices and ρ_i^- at even indices; there $\psi_i = \pm 1$, giving $\nabla g = \mathbf{G}^{(-1)^t}$. ■

Proof of Theorem 4. By Lemma 3 the normalized run produces the targets (34), so by Lemma 2 its estimator locks onto the wrong-sign cycle and $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t$ is a constant shift. Along it Φ_1, Φ_2 return to their values and the corrections b_k vanish, so only the linear term acts: $\tilde{f}(\tilde{\mathbf{X}}_{t+2}) - \tilde{f}(\tilde{\mathbf{X}}_t) = -\gamma(-\frac{7}{20}) = \frac{49}{240} > 0$. By Lemma 1, f then obeys (13) with $c = \frac{49}{240\tilde{L}}$. As $(L, \eta, \mu, \text{variant})$ were arbitrary, no rule $\eta = \eta(L, \mu)$ can prevent divergence. ■

Remark 1 (Boundedness below) *Only the linear term $-\gamma W_{22}$ makes $\tilde{f}$ unbounded below, and only as $W_{22} \to +\infty$ — a region the iterates never enter, since $(\tilde{\mathbf{X}}_t)_{22} \le \frac{7}{10}$ throughout (it decreases after the transient). Replacing $-\gamma W_{22}$ by any C^∞ function that agrees with it on $\{W_{22} \le 1\}$ and is constant on $\{W_{22} \ge 2\}$ therefore leaves the whole trajectory — and* (13) *— unchanged while making $\tilde{f}$ bounded below (Assumption 1); the theorem is stated with this modification in force.*

Remark 2 (Verification) *The construction is checked in two independent ways: symbolically, in exact rational arithmetic, and numerically, by running the float64 reference implementation of Algorithm 3 on the assembled $\tilde{f}$. Figure 4 confirms that at $\mu = 0$ EF21-SignMuon is the only one of the eight methods that diverges, the others (SignMuon, MuonUSign, MuonSign, EF21-MuonUSign, EF21-MuonSign, SignSGD, Muon) staying bounded; Figure 5 confirms that EF21-SignMuon diverges at the exact rate $\frac{49}{480}$ for every $\mu \in \{0, \frac{1}{2}, 0.9, 0.95, 0.99\}$ under both standard and Nesterov momentum, as the reduction predicts (`code/counterexamples/problems.py`, `code/counterexamples/plot_ef21_momentum.py`).*

A.7 Convergence of EF21-MuonUSign and EF21-MuonSign

We do not analyze the Error-Feedback methods from scratch. Instead we show that EF21-MuonUSign and EF21-MuonSign are *exact instances* of EF21-Muon (Gruntkowska et al. 2025, Algorithm 3), whose convergence is already established in the layer-wise, stochastic, federated setting. The whole task is to check that our algorithms meet the framework's hypotheses; the guarantees then transfer verbatim. Only one of these checks is non-trivial – that the scaled sign is a valid compressor (Lemma 4); the rest is bookkeeping. We keep the reduction in our own notation and defer the single change of variables to Table 4.

Throughout, the model is the tuple of matrix layers $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, with $d_i := m_i n_i$,

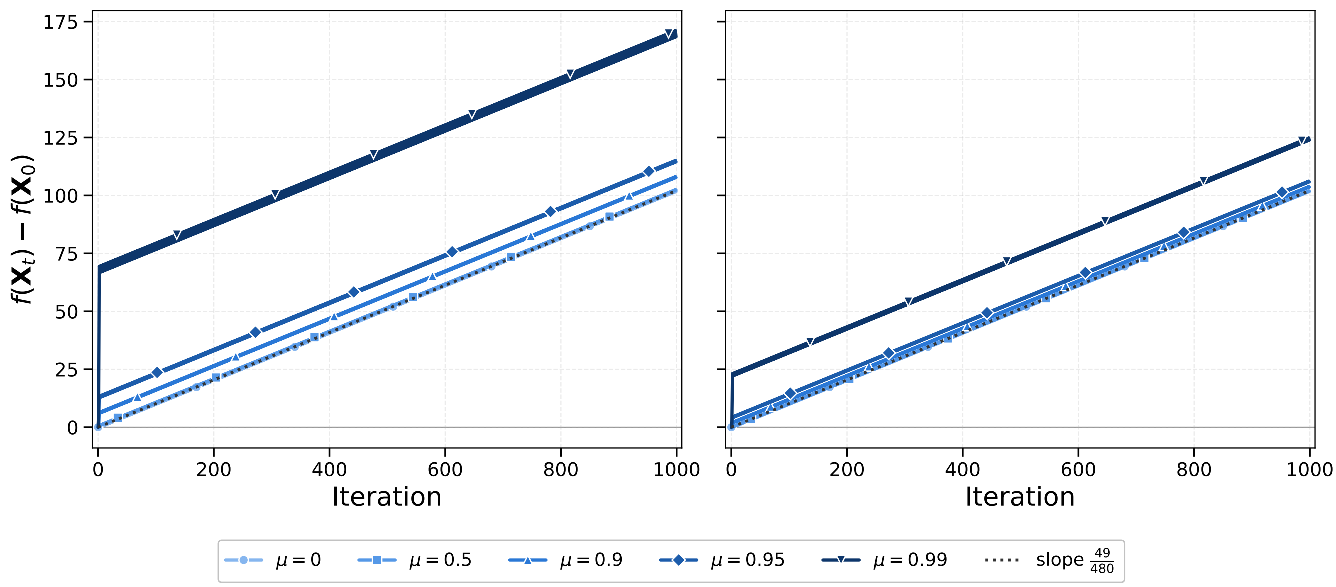


Figure 5: Momentum does not rescue EF21-SignMuon. For each momentum coefficient μ and variant, EF21-SignMuon is run on the corresponding instance of Theorem 4 and $f(\mathbf{X}_t) - f(\mathbf{X}_0)$ is plotted; *left:* standard momentum, *right:* Nesterov. Every setting diverges at the common rate $\frac{49}{480}$ (dotted). Subtracting $f(\mathbf{X}_0)$ removes the only genuinely μ-dependent offset (the field constant $A(\mu)$ scales a bounded periodic term); what remains is a bounded transient, largest as $\mu \to 1$, on top of the shared linear divergence.

$r_i := \min(m_i, n_i)$, $d_{\max} := \max_i d_i$; a second subscript selects a layer ($\mathbf{X}_{t,i}$, $\mathbf{G}_{t,i}$). We use $\|\mathbf{Y}\|_{2\to 2} \le \|\mathbf{Y}\|_F \le \|\mathbf{Y}\|_* \le \sqrt{\operatorname{rank} \mathbf{Y}}\, \|\mathbf{Y}\|_F$ between the spectral, Frobenius and nuclear norms.

Assumptions We use the standard assumptions of the framework, specialized to the spectral (Muon) geometry.

Assumption 1 (Lower boundedness) *$f(\mathbf{X}) \ge f^*$ for all $\mathbf{X}$, and (only for Corollary 2) each local $f_j \ge f_j^*$.*

Assumption 2 (Layer-wise smoothness) *Each layer carries the spectral norm $\|\cdot\|_{2\to 2}$, dual to the nuclear norm $\|\cdot\|_*$. For every layer i and all $\mathbf{X}, \mathbf{Y}$,*

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \le L_i^g\, \|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to 2}, \tag{43}$$

for $g = f$ (constant L_i) and $g = f_j$ (constant $L_{i,j}$); set $\tilde{L}_i^2 := \frac{1}{N}\sum_j L_{i,j}^2$ and $L := \max_i L_i$. The weaker layer-wise (L^0, L^1)-smoothness (Riabinin et al. 2025; Pethick et al. 2026) replaces L_i^g by $L^0 + L^1\|\nabla_i g(\mathbf{X})\|_*$; it reduces to the above when $L^1 = 0$.

Assumption 3 (Stochastic gradient) *Each client's stochastic gradient is unbiased, $\mathbb{E}_\xi[\nabla f_j(\mathbf{X};\xi)] = \nabla f_j(\mathbf{X})$, with bounded variance $\mathbb{E}_\xi\|\nabla f_j(\mathbf{X};\xi) - \nabla f_j(\mathbf{X})\|_*^2 \le \sigma^2$.*

These are Assumptions 1–2 and 6–10 of Gruntkowska et al. (2025) with the layer norms taken spectral; the variance bound is (mildly) stronger than theirs, which is stated in the Frobenius norm and implied by ours via $\|\cdot\|_F \le \|\cdot\|_*$.

Main result

Theorem 5 (Convergence of the EF21 methods) *Run the federated Algorithm 9 with the EF21 uplink and EMA momentum $\mu \in [0,1)$. Then both methods are exact instances of EF21-Muon (Proposition 2), and:*

(i) (smooth; both methods) *Under Assumptions 1–3, with the sharp learning rate $\eta_{t,i} = \gamma_i\|\mathbf{G}_{t,i}\|_*$ and suitably tuned (γ_i, μ), EF21-MuonUSign and the bidirectional EF21-MuonSign both reach*

$$\tfrac{1}{T}\textstyle\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$$

(Corollary 1).

(ii) (generalized smooth; uplink only) *Under the weaker (L^0, L^1)-smoothness, EF21-MuonUSign with a plain* constant *learning rate reaches $\min_{t\le T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4})$ (Corollary 2).*

By contrast, the majority-vote placements SignMuon, MuonUSign, MuonSign (Theorems 1–3) and EF21-SignMuon (Theorem 4) admit no such guarantee.

Part (i) gives the $\mathcal{O}(T^{-1/2})$ rate announced in the main text (via $\min_t \le$ average); the centralized Algorithms 6–7 are the case $N = 1$. The rest of this section proves the theorem by verifying the framework's four requirements.

The three routine checks

The step is a spectral LMO step. Writing $\mathbf{G} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, the framework's oracle over the spectral ball of radius τ is $\operatorname{lmo}(\mathbf{G}) = \mathbf{X} - \tau\mathbf{U}\mathbf{V}^\top$. Our server step is exactly this with $\tau = \eta_{t,i}$, since $\mathbf{D}_{t,i} = -A(\mathbf{G}_{t,i}) = \mathbf{U}_{t,i}\mathbf{V}_{t,i}^\top$:

$$\mathbf{X}_{t,i} = \mathbf{X}_{t-1,i} - \eta_{t,i}\,\mathbf{D}_{t,i} = \operatorname{lmo}_{\mathcal{B}(\mathbf{X}_{t-1,i},\eta_{t,i})}(\mathbf{G}_{t,i}). \tag{44}$$

The framework's "sharp" step $\mathbf{X} - \gamma \mathbf{G}^\sharp$ uses $\mathbf{G}^\sharp = \|\mathbf{G}\|_* \mathbf{U}\mathbf{V}^\top$ for the spectral norm; it is recovered by the schedule $\eta_{t,i} = \gamma_i \|\mathbf{G}_{t,i}\|_*$ (used in Corollary 1), while a constant $\eta_{t,i}$ is the plain Muon rate (Corollary 2). Both are learning-rate choices, not algorithmic changes. For the spectral norm the norm-equivalence constants are $\underline{\rho}_i = 1$, $\bar{\rho}_i = \sqrt{r_i}$. At rank-deficient $\mathbf{G}$ the LMO is non-unique; any selection works, since the analysis uses only $\langle \mathbf{G}, \mathrm{lmo}(\mathbf{G})\rangle = -\tau\|\mathbf{G}\|_*$ and $\|\mathrm{lmo}(\mathbf{G})\|_{2\to 2} \le \tau$.

Momentum and loop order. Algorithm 9 uses EMA momentum $\mathbf{M}_t = \mu \mathbf{M}_{t-1} + (1-\mu)\mathbf{G}_t$, which is the framework's momentum with $\beta = 1 - \mu$. Its heavy-ball form (Equation 7) gives *identical iterates*: with zero initialization one checks by induction that the heavy-ball states are the EMA states scaled by $1/(1-\mu)$, and both $\mathrm{sign}(\cdot)$ and the Muon LMO are invariant under positive scaling, so $\mathbf{X}_t$ is unchanged. (The optional Nesterov branch is not covered; see Remark 5.) The only remaining discrepancy is the order of operations within a round, which is resolved next.

Proposition 2 (Exact instance) *Fix μ, a learning-rate schedule, and the LMO selection above. Started from $\mathbf{M}_0^{(j)} = \mathbf{g}_0^{(j)} = \mathbf{G}_0 = \mathbf{0}$, $\mathbf{W}_0 = \mathbf{X}_0$, Algorithm 9 with the EF21 uplink and either downlink mode ($\mathcal{C}^{\downarrow} \in \{$exact, EF21-P$\}$) produces the same trajectory as Algorithm 3 of Gruntkowska et al. (2025) with spectral norms, scaled-sign worker compressors, identity/scaled-sign server compressor, $\beta = 1-\mu$, and radii $t_i^k = \eta_{k,i}$ – up to a one-step index shift $X^{t+1} = \mathbf{X}_t$.*

Proof The two loops differ only in where the round is cut: Gruntkowska et al. (2025) order it *step → downlink → gradient → uplink*, we order it *downlink → gradient → uplink → step*. Their iteration $k = 0$ is vacuous under our initialization: $\mathbf{G}_0 = \mathbf{0}$ gives $X^1 = X^0 = \mathbf{X}_0$ and a zero downlink residual, so $W^1 = \mathbf{X}_0$. Thereafter each of their iterations $k = t$ performs our round t verbatim – same momentum, same compressed residual added to the estimator, same LMO step (44) – giving $X^{t+1} = \mathbf{X}_t$, $W^{t+1} = \mathbf{W}_t$ by induction. Hence running their method for $K = T + 1$ iterations yields the iterates $\{\mathbf{X}_0, \mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_{T-1}\}$, and any average or minimum over them equals ours up to one duplicated (nonnegative) term. ■

Zero initialization also makes the framework's initial-error constant explicit: the $\|\mathbf{M}_0 - \mathbf{g}_0\|$ term vanishes and the gradient-deviation terms become $\|\nabla_i f(\mathbf{X}_0)\|$, so Ψ^0 is a finite constant.

The one real check: the scaled sign is a compressor The framework needs every transmitted message to come from a *contractive compressor*: a map $\mathcal{C}$ with $\mathbb{E}\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|^2 \le (1-\alpha)\|\mathbf{Y}\|^2$ for some $\alpha \in (0, 1]$ (Gruntkowska et al. 2025, Def. 1). This is exactly the property that fails for a bare sign and holds once it is scaled.

Lemma 4 (Contractivity of the scaled sign) *For every $\mathbf{Y} \in \mathbb{R}^{m\times n}$ ($d = mn$), the scaled sign $\mathcal{C}(\mathbf{Y}) = \mathrm{mean}(|\mathbf{Y}|)\,\mathrm{sign}(\mathbf{Y})$ satisfies*

$$\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - \tfrac{2}{d}\|\mathbf{Y}\|_1^2 + \tfrac{\|\mathbf{Y}\|_0}{d^2}\|\mathbf{Y}\|_1^2 \le \left(1 - \tfrac{1}{d}\right)\|\mathbf{Y}\|_F^2, \quad (45)$$

so $\mathcal{C}$ is contractive in the Euclidean norm with $\alpha = 1/d$. The bound $\alpha = 1/d$ is met only by 1-sparse inputs; for dense $\mathbf{Y}$ the true contraction is $\alpha(\mathbf{Y}) = \|\mathbf{Y}\|_1^2/(d\|\mathbf{Y}\|_F^2) = \Theta(1)$ (e.g. $\to 2/\pi$ for i.i.d. Gaussian entries).

Proof With $c := \|\mathbf{Y}\|_1/d$, zero entries contribute nothing and each nonzero entry contributes $(|Y_{kl}| - c)^2$, giving $\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + c^2\|\mathbf{Y}\|_0$, i.e. (45). Then $\|\mathbf{Y}\|_0 \le d$ and $\|\mathbf{Y}\|_1 \ge \|\mathbf{Y}\|_F$ give the bound; the dense case sets $\|\mathbf{Y}\|_0 = d$, and the Gaussian limit uses $\mathbb{E}|Y_{kl}| = \sqrt{2/\pi}\,\sigma$. ■

Remark 3 (Why the scaling is decisive) *The* unscaled *sign is not contractive for any α: as $\|\mathbf{Y}\|_F \to 0$ on a fixed support, $\|\mathrm{sign}(\mathbf{Y}) - \mathbf{Y}\|_F \to \sqrt{\|\mathbf{Y}\|_0}$ does not vanish. This is the exact line between our divergent and convergent methods. SignMuon, MuonUSign and MuonSign broadcast unscaled signs of full quantities – outside every error-feedback theory, and indeed divergent (Theorems 1–3). The EF21 variants instead send the* scaled *sign of a* residual*; the one extra scalar* $\mathrm{mean}(|\cdot|)$ *per layer per round, negligible on top of one bit per parameter, is what makes the compressor contractive.*

Applied per layer, the uplink scaled sign lies in $\mathbb{B}_2(1/d_i) \subseteq \mathbb{B}_2(\alpha)$ with the common parameter

$$\alpha := 1/d_{\max}. \quad (46)$$

The two methods differ only on the downlink: EF21-MuonUSign broadcasts the exact model (server compressor $= \mathcal{I}$, $\alpha_P = 1$), while EF21-MuonSign applies the same scaled sign, contributing a second contractive compressor in $\mathbb{B}_2(\alpha)$.

	This paper	**Gruntkowska et al. (2025)**
clients / rounds	N, T	$n, K = T + 1$
iterate	$\mathbf{X}_t$	X^{t+1}
momentum	μ	$1 - \beta$
learning rate	$\eta_{t,i}$	radius t_i^t
layer norm / dual	$\|\cdot\|_{2\to 2}, \|\cdot\|_*$	$\|\cdot\|_{(i)}, \|\cdot\|_{(i)\star}$
norm equivalence	$1,\ \sqrt{r_i}$	$\underline{\rho}_i,\ \bar{\rho}_i$
uplink compr.	scaled sign	$\mathbb{B}_2(\alpha_D)$
downlink compr.	exact / sign	$\mathcal{I}$ / $\mathbb{B}_2(\alpha_P)$
compression α	$1/d_{\max}$	α_D
smoothness	$L_i,\ \tilde{L}_i$	$L_i^0,\ \tilde{L}_i^0$

Table 4: Change of variables from our notation to that of Gruntkowska et al. (2025). The one-step index shift is Proposition 2.

Transferred guarantees All four requirements hold, so the framework's theorems apply through the change of variables of Table 4 ($\underline{\rho}_i = 1$, $\bar{\rho}_i = \sqrt{r_i}$, $\alpha_D = \alpha$). We state the resulting rates and stepsize rules; the explicit non-asymptotic bounds are those of Gruntkowska et al. (2025), Theorem 19 (smooth) and Theorem 24 ((L^0, L^1)), evaluated at these constants, and we do not reproduce them.

Corollary 1 (Smooth case; both methods) *Let Assumptions 1–3 hold. Run Algorithm 9 with the EF21 uplink, EMA momentum, and the sharp learning rate $\eta_{t,i} = \gamma_i\|\mathbf{G}_{t,i}\|_*$*

with any constants γ_i below the per-layer threshold of Gruntkowska et al. (2025, Thm. 19) under Table 4. Then, tuning μ as in Gruntkowska et al. (2025, Cor. 2),

$$\tfrac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2}).$$

Proof By Proposition 2 the run is an instance of Algorithm 3 for $K = T + 1$ iterations, and by Lemma 4 its compressors satisfy $\alpha_D = \alpha$ (uplink) and $\alpha_P \in \{1, \alpha\}$ (downlink; EF21-MuonSign's Euclidean downlink is admissible by their Remark 23, which adds a factor $\bar{\rho}_i^2 = r_i$ to the primal terms of the threshold). Substituting $\underline{\rho}_i = 1, \bar{\rho}_i = \sqrt{r_i}$ into Gruntkowska et al. (2025, Thm. 19) gives the stepsize threshold and the $\mathcal{O}(T^{-1/2})$ rate; the duplicated $\mathbf{X}_0$-term contributes only a $\frac{T+1}{T}$ factor. ■

The threshold shrinks polynomially in the layer dimension d_i (through $\bar{\rho}_i^2 = r_i$ and the uplink $\alpha = 1/d_{\max}$), and for EF21-MuonSign carries an extra r_i from the compressed downlink – the price of the spectral geometry and of one-bit messages (Remark 6). For the bidirectional method this penalty is structural: the scaled sign cannot satisfy the layer-norm contractivity that the unpenalized branch requires, for any layer of rank above one (Remark 7).

Corollary 2 ((L^0, L^1) case; EF21-MuonUSign) *Let Assumption 2 hold in its (L^0, L^1) form. Run EF21-MuonUSign (exact downlink) with $S := T+2$, momentum $\mu = 1 - S^{-1/2}$, and the* constant *per-layer rate $\eta_{t,i} \equiv \eta_i / S^{3/4}$ for any η_i small enough that $\eta_i \le 1$ and $\eta_i^2 \sqrt{r_i}\,(L^1_{i,\max})^2 = \mathcal{O}(1)$ (the explicit bound is Gruntkowska et al. (2025, Thm. 24)). Then*

$$\min_{0\le t\le T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4}).$$

Proof Their Theorem 24 requires the identity server compressor, which is exactly the exact downlink of EF21-MuonUSign (hence EF21-MuonSign is excluded here; Remark 5). Applying it with $K = T + 1$, $\beta = S^{-1/2}$ and the dictionary substitution yields the stated schedule and rate. ■

Remark 4 (The variant we actually run) *Under plain smoothness ($L^1 = 0$) the constraint in Corollary 2 collapses to $\eta_i \le 1$. So the* constant-learning-rate *EF21-MuonUSign used in our experiments – with no gradient-norm-dependent scaling – already converges at rate $\mathcal{O}(T^{-1/4})$; the sharp schedule of Corollary 1 is needed only to obtain the faster $\mathcal{O}(T^{-1/2})$ rate in squared norm.*

Scope and consequences

Remark 5 (What the reduction does and does not cover) *It covers the two Error-Feedback methods and no others, matching our negative results.* (i) *The majority-vote methods send unscaled signs and fall outside the framework (Remark 3); Theorems 1–3 confirm they diverge.* (ii) *EF21-SignMuon compresses the LMO* output *rather than the gradient, so it tracks the non-Lipschitz polar factor, the framework's momentum-tracking step breaks, and it diverges for every L-based step size (Theorem 4).* (iii) *EF21-MuonSign gets the smooth guarantee (Corollary 1) but not the (L^0, L^1) one, since the generalized-smooth theory of Gruntkowska et al. (2025) assumes an uncompressed downlink.* (iv) *The guarantees are for the default EMA branch; the Nesterov branch is not analyzed.*

Remark 6 (The price of one bit) *The uplink parameter $\alpha = 1/d_{\max}$ is what makes the stepsize threshold of Corollary 1 shrink with the dimension – the worst-case cost of one-bit messages, of the same nature as for Top-1 compressors in Euclidean EF21 (Richtárik, Sokolov, and Fatkhullin 2021).* On the uplink *it is a worst case only: it needs* 1*-sparse residuals, whereas the momentum residual is dense with $\alpha(\mathbf{\Delta}) = \Theta(1)$ (Lemma 4), and the framework admits iteration-dependent α (Gruntkowska et al. 2025, Rem. 12), recovering the constants realized in practice.*

The downlink of EF21-MuonSign does not inherit this reassurance. *Its residual is not a difference of gradients refreshed by fresh stochasticity each round, but the output of the compressor's own recursion*

$$\mathbf{\Delta}^{\downarrow}_{t+1} = \mathbf{\Delta}^{\downarrow}_t - \eta_t \mathbf{D}_t - \operatorname{mean}|\mathbf{\Delta}^{\downarrow}_t|\,\operatorname{sign}(\mathbf{\Delta}^{\downarrow}_t),$$

in which every coordinate is corrected by the same *amount. Coordinates driven persistently harder than that mean are never caught while the bulk holds the mean down, so the residual concentrates – and concentration is what drives α towards its $1/d$ floor. This positive feedback is absent from the uplink, and we observe it: $\alpha(\mathbf{\Delta}^{\downarrow})$ falls to 1.2×10^{-4} on the layers built from a zero initialization, four orders of magnitude below its own uplink on the same layers (Section 5.4). For the bidirectional method the dimension dependence of Corollary 1 is therefore realizable, not pessimistic.*

Remark 7 (Why the sharp route is closed) *Corollary 1 reaches EF21-MuonSign only through the $\bar{\rho}_i^2$-penalized branch of Gruntkowska et al. (2025, Rem. 23), and this is forced rather than convenient. The sharp branch asks for a server compressor in $\mathbb{B}(\alpha_P)$, contractive in the* layer *norm; by the norm-equivalence argument of Gruntkowska et al. (2025, App. C) a compressor in $\mathbb{B}_2(\alpha)$ lies there only if $\alpha > 1 - \underline{\rho}_i^2/\bar{\rho}_i^2 = 1 - 1/r_i$. The scaled sign has $\alpha \le 1$, with equality only on the trivial input, so for any layer of rank $r_i > 1$ it can* never *qualify – for our 768×3072 matrices the bar is $\alpha > 0.9987$ against an isotropic $2/\pi$. The factor $\bar{\rho}_i^2 = r_i$ is thus not removable by a sharper analysis of this compressor, only by changing it. Two admissible replacements are cheap: random dropout ($\mathcal{C}(\mathbf{Y}) = \mathbf{Y}$ with probability p, else $\mathbf{0}$) lies in $\mathbb{B}(p)$ for* every *norm, and the Top-K SVD compressor lies in $\mathbb{B}(1 - \sigma_{K+1}^2/\sigma_1^2)$ for the spectral norm (Gruntkowska et al. 2025, App. C). Both restore the sharp branch at a comparable budget, giving up the strictly one-bit downlink.*

Remark 8 (From Muon to Gluon) *Only the constants $(\underline{\rho}_i, \bar{\rho}_i) = (1, \sqrt{r_i})$ are spectral-norm-specific: Lemma 4 is a Euclidean statement and the framework's theorems hold for arbitrary layer norms. Replacing the Muon LMO with any per-layer norm – the Gluon setting (Riabinin et al. 2025), e.g. Scion's $\ell_1 \to \ell_\infty$ geometry for embeddings (Pethick et al. 2025) – gives* EF21-GluonUSign *and* EF21-GluonSign, *for which both corollaries hold with the corresponding $(\underline{\rho}_i, \bar{\rho}_i)$. Convergence is thus a property of error feedback plus scaled-sign compression, not of the spectral geometry.*

We assume the exact spectral LMO, as do all analyses of Muon-type methods (Li and Hong 2025; Kovalev 2025;

Riabinin et al. 2025; Gruntkowska et al. 2025); in practice it is the five-step Newton–Schulz approximation of Algorithm 1 (Amsel et al. 2026), and vector parameters and the last layer are trained with AdamW, as is standard for Muon (Jordan et al. 2024). Finally, we note that although SignMuon carries no worst-case guarantee, empirically it stays within 1–1.5% of full-precision Muon while cutting uplink communication $\sim 32\times$ and beating SignSGD, in both the centralized and federated regimes.

A.8 Reproducibility Details

Computing infrastructure. The experiments were run on **[GPU model and memory, e.g. NVIDIA A100 80 GB]**, **[CPU model]**, with **[amount of RAM]** of system memory, under **[operating system, e.g. Ubuntu 22.04]**. The software stack was Python 3.9, **[PyTorch version]**, and **[CUDA/cuDNN versions]**; exact package versions are pinned in `code/requirements.txt`. Where experiments were executed on more than one machine, the per-experiment configuration is **[specify or state "identical across all experiments"]**.

Randomness and seeds. All experiments use a single fixed random seed (default 0), set through a `seed_everything` routine that seeds the Python `random` module, NumPy, and PyTorch (including all CUDA devices) and enables deterministic cuDNN kernels (`cudnn.deterministic = True`, `cudnn.benchmark = False`). Each reported number corresponds to a single such run.

A.9 Results for the smooth convex problem

Table 5 details the grid-search ranges used for each optimizer on the synthetic problem of Equation 18, and Figure 6 shows the loss and gradient-norm dynamics.

A.10 Centralized training results

Table 6 reports the centralized CIFAR-10 results on ResNet-18, including the tuned hyperparameters and the final train/test accuracies.

A.11 Federated training results. Experiment 1

Table 7 and Figure 7 report the comparison of optimizers with $N = 3$ clients (Experiment 1).

A.12 Federated training results. Experiment 2

Figure 8 (together with Table 2 in the main text) reports the comparison of optimizers with $N = 10$ clients (Experiment 2).

A.13 Language-modelling details

Setup. Upstream modded-nanoGPT record #40 (2025-10-04), the last record before NorMuon and hence the last whose hidden-matrix optimizer is a clean, separable momentum $\to$ LMO $\to$ step Muon, so our variants inject exactly at the LMO. Model: 12 layers, model dimension 768, 6 heads of dimension 128, vocabulary 50,257; hidden matrices are 768×3072 (the merged $\mathbf{Q}/\mathbf{K}/\mathbf{V}/\mathbf{O}$ weight is used as four 768×768 blocks, and both the LMO and the compressor scale are applied per block). Data: FineWeb10B, 262,144 tokens per step, 2330 steps (= 611M tokens), validation on the fixed 10,485,760-token split. Hardware: $8\times$H100, one process per GPU; gradients are averaged by `reduce_scatter` so the owning rank runs the centralized update and `all_gather` returns the parameter, i.e. the compression is a property of the update rule, as in the centralized algorithms we analyze.

Hyperparameters. Matrix/gate optimizer: $\eta_0 = 0.06$ (LMO family) or 0.03 (SIGN family), per-layer scaled by the unit-gain rule (Appendix A.14); Nesterov momentum $\mu = 0.95$, warmed up linearly from 0.85 over the first 300 steps and cooled back to 0.85 over the last 50; weight decay 0 (the record's own value); η constant then linearly cooled to $0.1\eta_0$ over the final 45% of the run; LMO by 5 Polar-Express iterations. Auxiliary parameters (embeddings, scalars, head) use the record's distributed Adam unchanged: $\eta = 0.008$, $\beta = (0.65, 0.95)$, $\varepsilon = 10^{-8}$, no weight decay, per-parameter multipliers 75 on embeddings and 5 on scalars, stepped every other iteration. Nothing above was tuned by us: the LMO family runs at the record's own η_0, and every value outside the matrix optimizer is the record's.

Compressor diagnostics. Table 8 reports, per layer type at the final step, the contraction each scaled sign achieves, $\alpha(\mathbf{\Delta}) = \|\mathbf{\Delta}\|_1^2/(d\|\mathbf{\Delta}\|_F^2)$, and the relative estimator lag. Two points are worth isolating. *(i)* Every uplink is well-contractive, $\alpha \in [0.32, 0.64]$ against the isotropic $2/\pi = 0.637$, and EF21-SignMuon's is uniformly the best because its target is orthogonal; yet the uplink *lag* is large (0.36–0.82), meaning the estimator is far from its target at every step and the methods nonetheless train well – the LMO is scale-invariant and needs only the direction. *(ii)* The single anomaly in the table is the `c_proj` downlink, $\alpha = 1.2\times10^{-4}$, four orders of magnitude below its uplink on the very same layer. Since both compressors are the same operator, the difference is a property of the residual, not of the compressor: the uplink residual is refreshed by an exogenous stochastic gradient each round, the downlink residual is generated by the compressor's own recursion (Remark 6).

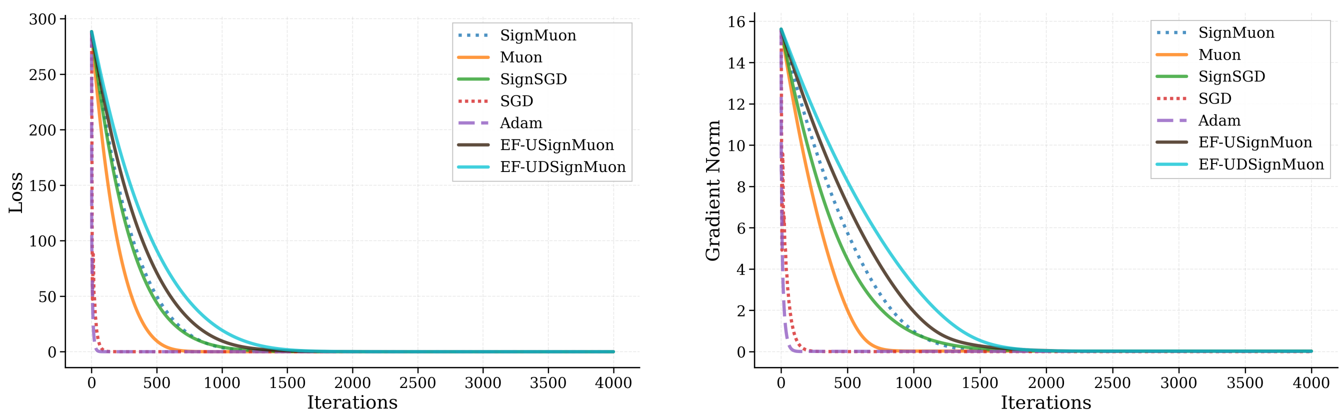
A.14 Per-Layer Step Sizes: the Unit-Gain Rule

Our counterexamples, like the analysis of Mishra, Trivedi, and Kumar (2026), concern a single matrix, where one scalar step size suffices. A network has layers of very different shapes, and the methods of this paper produce step matrices from *two* families whose norms scale differently with shape, so a single global η cannot be simultaneously correct for both families and across layers. This is not a gap we must tolerate: the layer-wise LMO framework to which our convergence result reduces (Riabinin et al. 2025; Gruntkowska et al. 2025) already carries per-layer norms $\|\cdot\|_{(\ell)}$, smoothness constants L_ℓ and radii η_ℓ — it is only the *experiments* that have been setting that radius to a constant. What follows instantiates it.

The two families. For a parameter reshaped to $\mathbf{X} \in \mathbb{R}^{m\times n}$ (m = fan-out, n = fan-in, $r = \mathrm{rank} = \min(m, n)$ generi-

Algorithm	Learning Rate (η)	Momentum (μ)
SignMuon	$[1 \cdot 10^{-4}, 1 \cdot 10^{-3}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
MuonUSign	$[1 \cdot 10^{-4}, 1 \cdot 10^{-3}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
EF21-MuonUSign	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
EF21-MuonSign	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Muon	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-3}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SignSGD	$[5 \cdot 10^{-5}, 2 \cdot 10^{-4}]$ with step $1 \cdot 10^{-5}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
SGD	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	$\{0.1, 0.2, \ldots, 0.9, 0.95\}$
Adam	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	–

Table 5: Hyperparameter search grid for the synthetic convex problem.



(a) Loss

(b) Frobenius norm of the gradient

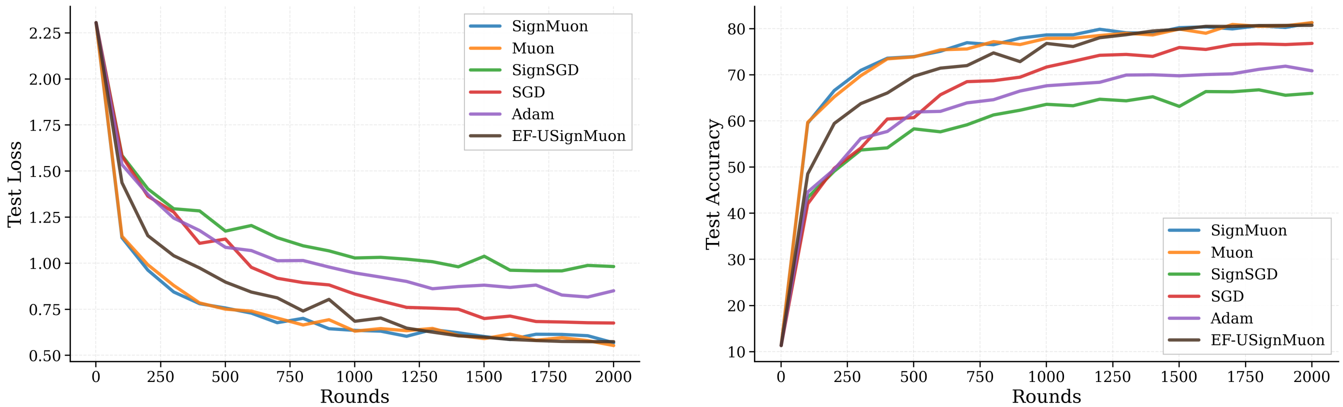
Figure 6: Smooth convex problem (Equation 18 for 500×500).

Dataset	Optimizer	Epoch	Batch	LR	LR Aux	Mom	Train Acc	Test Acc
CIFAR-10	SignMuon	75	128	0.001	0.0001	0.9	99.69%	92.55%
	MuonUSign			-	-		-%	-%
	Muon			0.015	0.001		99.94%	93.70%
	SGD						99.98%	93.93%
	SignSGD			0.001	0.0001		93.19%	89.11%
	Adam						99.77%	93.03%
	EF21-MuonUSign			0.008	0.001		99.96%	93.82%
	EF21-MuonSign			0.007	0.001		99.95%	93.86%

Table 6: CIFAR-10: Centralized Learning on ResNet-18.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	3	1	64	0.001	0.9	81.24%
	MuonUSign					-		-%
	Muon					0.05		81.27%
	SignSGD					0.001		65.96%
	SGD					0.03		76.77%
	Adam					0.001		70.83%
	EF21-MuonUSign					0.02		80.69%
	EF21-MuonSign					0.022		80.57%

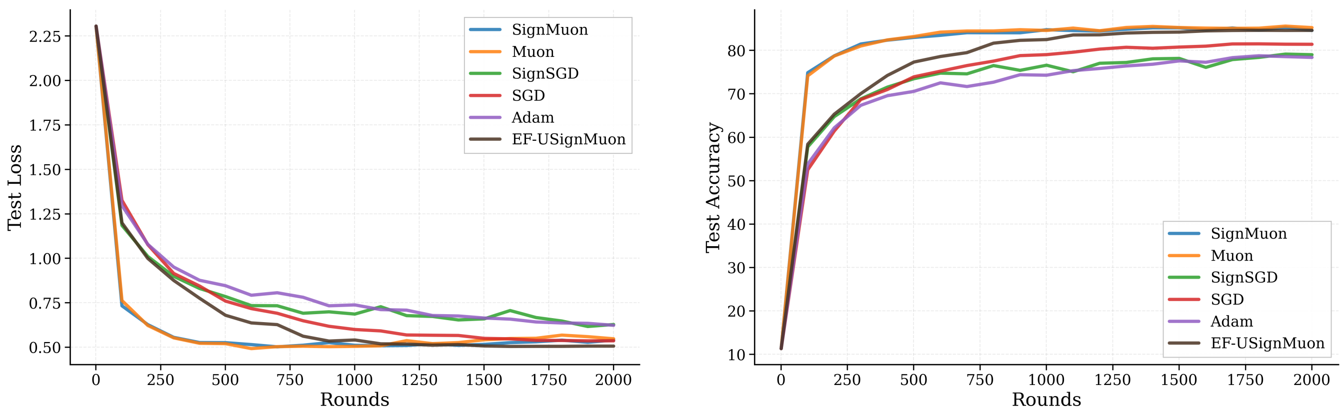
Table 7: CIFAR-10 Federated Learning on CNN2. Experiment 1: Comparison of Optimization Algorithms with (N = 3) Clients.



(a) Test Loss

(b) Test Accuracy

Figure 7: CIFAR-10 Federated Learning on CNN2. Experiment 1: Comparison of Optimization Algorithms with (N = 3) Clients.



(a) Test Loss

(b) Test Accuracy

Figure 8: CIFAR-10 Federated Learning on CNN2. Experiment 2: Comparison of Optimization Algorithms with (N = 10) Clients.

cally), the step matrices are

$$\begin{aligned} &\text{LMO family (Muon, MuonUSign, EF21-Muon\{USign, Sign\}, EF21-SignMuon)}: && \mathbf{s} = \mathbf{U}\mathbf{V}^\top, \\ &\text{SIGN family (SignMuon, MuonSign, SignSGD)}: && \mathbf{s} \in \{\pm 1\}^{m\times n}. \end{aligned} \tag{47}$$

Both have *exactly* known Frobenius norms: $\|\mathbf{U}\mathbf{V}^\top\|_F^2 = \operatorname{tr}(\mathbf{V}\mathbf{U}^\top\mathbf{U}\mathbf{V}^\top) = \operatorname{tr}(\mathbf{V}^\top\mathbf{V}) = r$, so $\|\mathbf{s}\|_F = \sqrt{r}$; and a ± 1 matrix has $\|\mathbf{s}\|_F = \sqrt{mn}$. (EF21-SignMuon steps along the error-feedback estimator $\mathbf{d}_t^{\mathrm{est}}$ of $\operatorname{polar}(\tilde{\mathbf{M}}_t)$ rather than along the oracle output itself, so for it $\|\mathbf{s}\|_F = \sqrt{r}$ holds only in the limit; we assign it to the LMO family on that basis.)

The criterion. Define the *RMS gain* of $\mathbf{A} \in \mathbb{R}^{m\times n}$, i.e. how much it amplifies a generic input in root-mean-square terms, with $\operatorname{rms}(\mathbf{v}) := \|\mathbf{v}\|/\sqrt{\dim}$:

$$\gamma(\mathbf{A})^2 := \frac{\mathbb{E}_{\mathbf{u}}\left[\operatorname{rms}(\mathbf{A}\mathbf{u})^2\right]}{\mathbb{E}_{\mathbf{u}}[\operatorname{rms}(\mathbf{u})^2]}, \qquad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \tag{48}$$

Since $\mathbb{E}\|\mathbf{A}\mathbf{u}\|^2 = \operatorname{tr}(\mathbf{A}^\top\mathbf{A}) = \|\mathbf{A}\|_F^2$ and $\mathbb{E}\|\mathbf{u}\|^2 = n$, (48) evaluates in closed form:

$$\gamma(\mathbf{A}) = \frac{\|\mathbf{A}\|_F}{\sqrt{m}}. \tag{49}$$

No random-matrix theory and no distributional fitting enter; the single modelling assumption is that $\mathbf{u}$ is isotropic *and independent of* $\mathbf{A}$, which we return to below.

Every standard fan-in-scaled initialization has *shape-independent* gain. For He normal ($\sigma^2 = 2/n$), $\|\mathbf{X}\|_F = \sigma\sqrt{mn} = \sqrt{2m}$, so $\gamma(\mathbf{X}) = \sqrt{2}$; for PyTorch's default Kaiming-uniform convolution ($\sigma^2 = 1/(3n)$), $\gamma(\mathbf{X}) = 1/\sqrt{3}$. Either way a constant. Requiring the update's gain to be a fixed fraction of the weight's is therefore simply the requirement that *the per-step gain be the same on every layer*, and by (49) that is one formula:

$$\boxed{\eta_\ell = \eta_0\,\lambda_\ell, \qquad \lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{s}\|_F}} \tag{50}$$

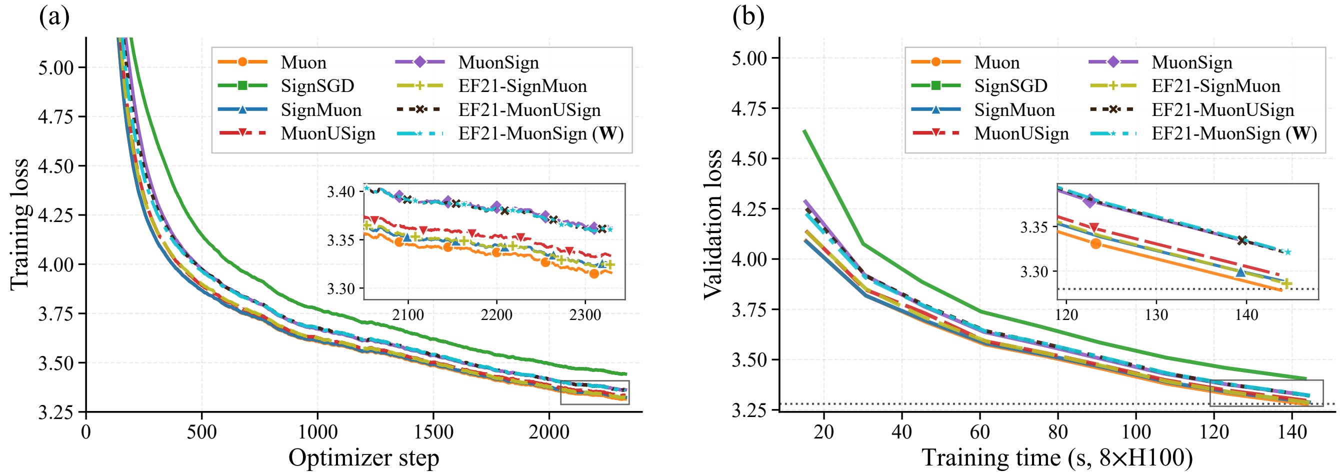


Figure 9: NanoGPT, supporting curves. **(a)** Training loss against optimizer step (EMA-smoothed; the logged quantity is a per-rank sum over 32,768 tokens, divided out here). **(b)** Validation loss against the speedrun clock, which excludes validation and compilation. Insets magnify the boxed tails, as in Figure 3. The ordering is the same on both axes: no method pays a measurable wall-clock premium for its compressor or its error-feedback buffers. Note in (a) that EF21-MuonSign's training loss – taken at $\mathbf{W}$ – is healthy throughout, which is what isolates its validation gap to the $\mathbf{X}$-tracking rather than to training.

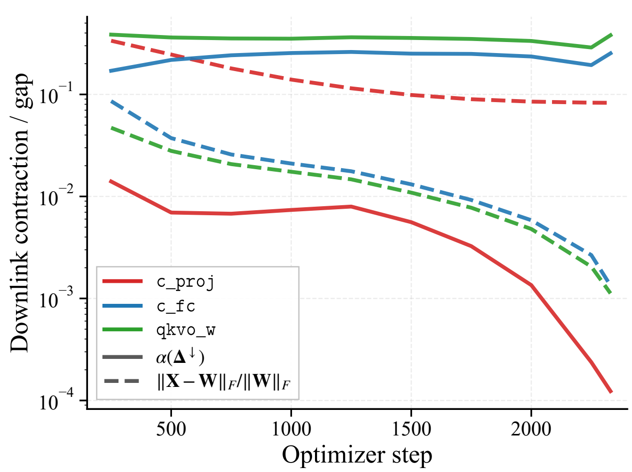


Figure 10: The downlink measurement for EF21-MuonSign, per layer type: the contraction $\alpha(\mathbf{\Delta}^{\downarrow})$ the scaled sign actually achieves (solid) and the resulting server/broadcast gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$ (dashed). On the zero-initialized `c_proj` the contraction collapses by three orders of magnitude and the gap stalls near 10^{-1}; every other layer type contracts at $\Theta(10^{-1})$ and closes its gap by $\sim 65\times$ over the run. This is Remark 6 measured.

	uplink		downlink	
Layer type	α	lag	α	gap
EF21-SignMuon				
`qkvo_w` ($\times 10$)	0.640	0.66	–	–
`c_fc` ($\times 11$)	0.633	0.62	–	–
`c_proj` ($\times 11$)	0.634	0.62	–	–
EF21-MuonUSign				
`qkvo_w`	0.374	0.81	–	–
`c_fc`	0.323	0.82	–	–
`c_proj`	0.597	0.62	–	–
EF21-MuonSign				
`qkvo_w`	0.406	0.79	0.380	0.0011
`c_fc`	0.355	0.78	0.253	0.0013
`c_proj`	0.596	0.61	$\mathbf{1.2\times10^{-4}}$	**0.083**

Table 8: Compressor diagnostics at step 2330, medians over the identical layers of each type. α is the contraction the scaled sign achieves on that round's residual ($2/\pi$ for an isotropic residual, $1/d$ in the worst case); "lag" is $\|\text{target} - \text{estimator}\|_F/\|\text{target}\|_F$; "gap" is $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$. Downlink columns are empty for the two methods that broadcast exactly. Gates are omitted (they are 6×12 and 1×12).

which gives $\gamma(\eta_\ell \mathbf{s}) = \eta_0$ exactly, for every shape and both families – so η_0 *is* the per-step RMS gain. Substituting the two Frobenius norms of (47),

$$\lambda_\ell^{\text{LMO}} = \sqrt{\max\left(1, \tfrac{m}{n}\right)}, \qquad \lambda_\ell^{\text{SIGN}} = \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{\text{fan-in}}}. \tag{51}$$

Why we believe (51)**.** The first expression is exactly the aspect-ratio factor $\sqrt{\max(1, m/n)}$ present in the reference Muon implementation (Jordan et al. 2024), which was introduced as a practical heuristic. The unit-gain criterion *derives* it, and also explains why Muon's step size is known to transfer across widths: its step has $\gamma = \eta$ independently of n, so no fan-in correction is needed. The second expression is the counterpart the sign family has never been given. Only η_0 is tuned, and it is now a shape-free quantity; the shape dependence is fixed a priori.

Two consequences are worth recording. First, λ_ℓ is a deterministic function of the layer shape, known to server and clients alike, so per-layer step sizes cost *nothing* in communication and leave the one-bit-per-parameter budget untouched. Second, on the CIFAR ResNet-18 of our experiments $\lambda_\ell^{\text{SIGN}}$ spans a factor of 13 across layers (from $1/\sqrt{27}$ at the stem to $1/\sqrt{4608}$ in the last stage), so a single global rate is necessarily a compromise: roughly correct for the middle of the network, several-fold too large at the stem and too small at the end. The LMO family is spared this, which is one reason full-precision Muon is easier to tune than its sign-compressed variants.

The one open exponent. Writing $\lambda_\ell^{\text{SIGN}} = n^{-\alpha}$, the unit-gain rule is $\alpha = \frac{1}{2}$. Identity (49) requires the input to be independent of $\mathbf{A}$, which is the right model for a *single* step; if the *accumulated* update $\sum_t \eta \mathbf{s}_t$ correlates with the activations, its gain can be as large as $\|\mathbf{A}\|_{\text{op}}\sqrt{n/m} = \Theta(\eta n)$ rather than $\Theta(\eta\sqrt{n})$, giving $\alpha = 1$ – the classical μP rule $\eta \propto 1/\text{fan-in}$ for sign-like (Adam-type) updates (Yang et al. 2021). Thus $\alpha \in [\frac{1}{2}, 1]$ interpolates between an incoherent and a fully aligned accumulation, and the endpoint is an empirical question rather than a modelling choice. Two measurements bear on it. At the single-step level we find $\|\operatorname{sign}(\operatorname{polar}(\mathbf{M}))\|_{\text{op}} = 0.93\,(\sqrt{m} + \sqrt{n})$ to within 8% at every ResNet-18 layer shape, i.e. a single sign step is spectrally *incoherent*, not rank-one aligned; and we track the realized gain $\|\mathbf{X}_t - \mathbf{X}_0\|_F/\sqrt{m}$ during training, whose growth exponent in t separates the two regimes directly. Fitting that exponent on CIFAR ResNet-18 at a *constant* step size – annealing would make the accumulation saturate and the fit would measure the schedule instead – gives 0.28 for Muon and 0.12 for SignMuon over 20 epochs ($R^2 \geq 0.87$). Both are far below 1: the accumulated update does not align, and indeed grows sub-diffusively, so successive steps partially cancel. Nothing in the measurement warrants raising α above the value $\frac{1}{2}$ that the exact single-step identity (49) already delivers, and we adopt it.

Weight decay is not free to place. The same scale invariance that makes λ_ℓ necessary also dictates *where* an ℓ_2 penalty may be applied. Every step map considered here is positively homogeneous of degree *zero*: $\operatorname{sign}(c\mathbf{M}) = \operatorname{sign}(\mathbf{M})$ and $\operatorname{polar}(c\mathbf{M}) = \operatorname{polar}(\mathbf{M})$ for all $c > 0$. Consequently, folding a decay term into the gradient before the LMO – $\widetilde{\mathbf{G}}_t = \mathbf{G}_t + \lambda_{\text{wd}}\mathbf{W}_t$, the convention of Mishra, Trivedi, and Kumar (2026) and of most sign-method implementations – cannot shrink $\mathbf{W}_t$ at all: the step length $\eta_t\lambda_\ell\|\mathbf{s}\|_F$ is fixed by (47), and the decay term acts only by *rotating* the direction. For the sign-terminated steps the cancellation is exact to machine precision; for the LMO-terminated ones it is exact for the polar factor itself, and the few percent by which a Newton–Schulz approximation of it moves is the iteration's error, not shrinkage. The rotation is governed by the ratio $\rho_t = \lambda_{\text{wd}}\|\mathbf{W}_t\|_F/\|\mathbf{G}_t\|_F$, and it is not benign: averaged over the ResNet-18 layer shapes, $\rho = 0.1$ already flips 3.4% of the coordinates of $\operatorname{sign}(\operatorname{polar}(\cdot))$, and $\rho = 1$ flips 26%, the polar direction retaining cosine similarity only 0.68 with the undecayed one. Since $\|\mathbf{G}_t\|_F$ falls by orders of magnitude over training while $\|\mathbf{W}_t\|_F$ grows, ρ_t drifts from negligible to $\Theta(1)$; worse for a comparison, it depends on each method's own momentum scale, so one nominal λ_{wd} is a different perturbation for each. Decoupled decay, $\mathbf{W}_{t+1} = (1 - \eta_t\lambda_{\text{wd}})\mathbf{W}_t - \eta_t\lambda_\ell\mathbf{s}_t$, is by contrast exactly commensurate with the update under the unit-gain rule: the displacement it contributes has gain $\eta_t\lambda_{\text{wd}}\gamma(\mathbf{W}_t)$ against the step's gain η_t, so their ratio is $\lambda_{\text{wd}}\gamma(\mathbf{W}_t)$ – free of η_t, of the layer shape, and of the method. We therefore decouple, and use the coupled form only for an ablation. This also explains an otherwise puzzling observation of Mishra, Trivedi, and Kumar (2026): sweeping $\lambda_{\text{wd}} \in \{0, 0.1, 0.2\}$ coupled, all ten of their best CIFAR-10 configurations select $\lambda_{\text{wd}} = 0$. At their two nonzero values $\rho_t \gg 1$ throughout training, so the update degenerates towards $-\operatorname{sign}(\mathbf{W}_t)$; the sweep rejects the placement rather than the regularizer.

What the analysis covers. Our theorems are stated for unregularized f, so we report unregularized runs as the primary comparison and weight decay as an ablation; the reference nanoGPT configuration we build on also uses $\lambda_{\text{wd}} = 0$ for every parameter group. Two remarks delimit the gap. First, the *coupled* form is covered verbatim: it is nothing other than running the same method on $f_\lambda = f + \frac{\lambda_{\text{wd}}}{2}\|\mathbf{W}\|_F^2$, which is $(L + \lambda_{\text{wd}})$-smooth, so every rate carries over with $L \mapsto L + \lambda_{\text{wd}}$. The irony is that this is exactly the variant that does not regularize. Second, the *decoupled* form is not covered by our rates, but it buys something the analysis wants. Since $\|\mathbf{s}_t\|_F$ is a known constant and $\lambda_\ell\|\mathbf{s}_t\|_F = \sqrt{m}$ identically under (50), the triangle inequality gives $\|\mathbf{W}_{t+1}\|_F \leq (1 - \eta_t\lambda_{\text{wd}})\|\mathbf{W}_t\|_F + \eta_t\sqrt{m}$, and hence, for any $\eta_t\lambda_{\text{wd}} \leq 1$,

$$\gamma(\mathbf{W}_t) \;\leq\; \max\bigl\{\gamma(\mathbf{W}_0),\ \lambda_{\text{wd}}^{-1}\bigr\} \quad \text{for all } t, \tag{52}$$

a bound on the layer's gain that is uniform in t and *independent of the layer shape*. Norm-constrained updates are what make this possible: for SGD the step length is data-dependent and no such a priori bound exists. Since layer-wise LMO analyses assume smoothness on a bounded region, (52) is the statement that decoupled decay supplies that region rather than complicating it. A convergence rate for the decoupled variant itself we do not claim.

Comparison with concurrent work. Mishra, Trivedi, and Kumar (2026) analyse the normalized update $\mathbf{D}_t = \bar{\mathbf{S}}_t/\sqrt{mn}$, justified by $\|\mathbf{D}_t\|_{\text{op}} \leq \|\mathbf{D}_t\|_F = 1$ under their spectral-norm smoothness assumption, and remark that updating with $\bar{\mathbf{S}}_t$ directly is equivalent after absorbing $\sqrt{mn}$ into η_t. That equivalence holds for a single matrix but not across layers of differing shape, and their algorithm applies no shape factor, so their experiments use a single global rate as well. The substitution is also loose in a shape-dependent way: since $\|\operatorname{sign}(\operatorname{polar}(\mathbf{M}))\|_{\text{op}} \approx \sqrt{m} + \sqrt{n}$, the bound $\sqrt{mn}$ overstates the operator norm by a factor growing like $\sqrt{\min(m,n)}$ – from 3.7× at the stem to 18.1× in the last stage of a ResNet-18 – so the induced spectral trust region drifts roughly fivefold across the network instead of remaining flat.

Rule	LMO	SIGN	Equalizes
global ($\alpha = 0$)	1	1	nothing
Muon default	$\sqrt{\max(1, \frac{m}{n})}$	1	LMO gain only
unit gain	$\sqrt{\max(1, \frac{m}{n})}$	$n^{-1/2}$	per-step gain, both families
μP ($\alpha = 1$)	$\sqrt{\max(1, \frac{m}{n})}$	n^{-1}	gain of the aligned accumulation
Mishra, Trivedi, and Kumar (2026)	$r^{-1/2}$	$(mn)^{-1/2}$	$\|\mathbf{s}\|_F$ (the analysis)

Table 9: Per-layer step-size multipliers λ_ℓ. Only η_0 is tuned; λ_ℓ is fixed a priori by the shape. The rules differ only in the SIGN family, and the LMO column of the unit-gain rule coincides with the factor already used in practice (Jordan et al. 2024) – which is our main evidence that (48) is the right criterion.

Algorithm 1: MuonLMO

Input: tensor $\mathbf{Y}$; Newton-Schulz coefficients $a = 3.4445$, $b = 4.7750$, $c = 2.0315$; iteration count ns_steps
Output: Muon LMO direction $\mathbf{D} \approx \mathbf{U}\mathbf{V}^\top$ of $\mathbf{Y}$

$\mathbf{Y} \leftarrow \text{ReshapeTo2D}(\mathbf{Y})$ ▷ Flatten tensor to matrix $m \times n$
$\mathbf{Y} \leftarrow \mathbf{Y}/\|\mathbf{Y}\|_F$ ▷ Normalize (optional)
for $k = 1$ to ns_steps **do** ▷ 5th-order Newton-Schulz orthogonalization
  $\mathbf{A} \leftarrow \mathbf{Y}\mathbf{Y}^\top$
  $\mathbf{Y} \leftarrow a\,\mathbf{Y} - b\,\mathbf{A}\mathbf{Y} + c\,\mathbf{A}^2\,\mathbf{Y}$
end for
$\mathbf{D} \leftarrow \text{ReshapeToOriginal}(\mathbf{Y})$ ▷ Reshape back to the original tensor shape
return $\mathbf{D}$

A.15 Algorithms

All six federated methods are instances of just two templates, separated by *where the Muon LMO is evaluated*. When the sign acts *after* the LMO (the SignMuon family), each client must orthogonalize locally, so the LMO runs on the worker and the client transmits a compressed *direction* (Algorithm 8). When the sign acts *before* the LMO (the MuonUSign/MuonSign family), the client transmits a compressed *gradient*, and the server reconstructs it and applies a single LMO (Algorithm 9). Within each template, a method is fixed by its uplink compressor $\mathcal{C}^\uparrow \in \{\text{sign}, \text{EF21}\}$ and downlink compressor $\mathcal{C}^\downarrow \in \{\text{exact}, \text{sign}, \text{EF21-P}\}$; Table 10 lists the six instantiations. As in the centralized setting, both templates are applied per matrix parameter, while vector parameters and the final classification layer are optimized with AdamW.

Algorithm 2: SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu\mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu\mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\mathbf{s}_t^\uparrow \leftarrow \text{sign}(\mathbf{D}_t)$ ▷ Uplink sign compression
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^\uparrow$ ▷ Update parameters
end for

Algorithm 3: EF21-SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{d}_0^{\text{est}} \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu\mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation (EMA)
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu\mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\Delta_t^\uparrow \leftarrow \mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}$ ▷ Uplink residual (LMO direction)
  $\alpha_t^\uparrow \leftarrow \text{mean}(|\Delta_t^\uparrow|)$
  $\mathbf{d}_t^{\text{est}} \leftarrow \mathbf{d}_{t-1}^{\text{est}} + \alpha_t^\uparrow\,\text{sign}(\Delta_t^\uparrow)$ ▷ Uplink EF21
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{d}_t^{\text{est}}$ ▷ Update parameters
end for

Algorithm 4: MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t

Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\tilde{\mathbf{M}}_t)$ ▷ Uplink sign compression
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{s}_t^{\uparrow})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters
end for

Algorithm 5: MuonSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t

Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\mathbf{s}_t^{\uparrow} \leftarrow \mathrm{sign}(\tilde{\mathbf{M}}_t)$ ▷ Uplink sign compression
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{s}_t^{\uparrow})$
  $\mathbf{s}_t^{\downarrow} \leftarrow \mathrm{sign}(\mathbf{D}_t)$ ▷ Downlink sign compression
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\downarrow}$ ▷ Update parameters
end for

Algorithm 6: EF21-MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t

Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\mathrm{est}} \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\mathrm{est}}$ ▷ Uplink residual
  $\alpha_t^{\uparrow} \leftarrow \mathrm{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{g}_t^{\mathrm{est}} \leftarrow \mathbf{g}_{t-1}^{\mathrm{est}} + \alpha_t^{\uparrow}\,\mathrm{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{g}_t^{\mathrm{est}})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters
end for

Algorithm 7: EF21-MuonSign

Input: Initial model $\mathbf{X}_0 = \mathbf{W}_0$, momentum coefficient μ, learning rate η_t

Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\mathrm{est}} \leftarrow 0$
for $t = 1$ to T **do**
  Stochastic gradient $\mathbf{G}_t$ at the broadcast model $\mathbf{W}_{t-1}$
  $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
  $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
  $\Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\mathrm{est}}$ ▷ Uplink residual
  $\alpha_t^{\uparrow} \leftarrow \mathrm{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{g}_t^{\mathrm{est}} \leftarrow \mathbf{g}_{t-1}^{\mathrm{est}} + \alpha_t^{\uparrow}\,\mathrm{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{g}_t^{\mathrm{est}})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters (server)
  $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$ ▷ Downlink residual
  $\alpha_t^{\downarrow} \leftarrow \mathrm{mean}(|\Delta_t^{\downarrow}|)$
  $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow}\,\mathrm{sign}(\Delta_t^{\downarrow})$ ▷ Downlink EF21-P
end for

Method	**LMO**	**Uplink $\mathcal{C}^{\uparrow}$**	**Downlink $\mathcal{C}^{\downarrow}$**
SignMuon	worker	sign / MV	exact
EF21-SignMuon	worker	EF21	exact
MuonUSign	server	sign / MV	exact
MuonSign	server	sign / MV	sign
EF21-MuonUSign	server	EF21	exact
EF21-MuonSign	server	EF21	EF21-P

Table 10: The six federated methods as instantiations of the two templates: Algorithm 8 (worker-side LMO; rows 1–2) and Algorithm 9 (server-side LMO; rows 3–6). Each method is fixed by the LMO location and the uplink/downlink compressors. "MV": majority vote; "EF21-P": primal (model-side) error feedback; "exact": full-precision broadcast.

Algorithm 8: Federated SignMuon / EF21-SignMuon (worker-side LMO)

Input: initial model $\mathbf{X}_0$; clients N; rounds T; learning rate η_t; momentum μ; uplink compressor $\mathcal{C}^{\uparrow} \in \{\mathrm{sign}, \mathrm{EF21}\}$ (Table 10)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{d}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{d}_0 \leftarrow 0$
for $t = 1$ to T **do**
  broadcast $\mathbf{X}_{t-1}$ to all clients ▷ exact downlink
  for $j = 1$ to N **in parallel do** ▷ client j
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{X}_{t-1};\, \xi_t^{(j)})$
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\, \mathbf{G}_t^{(j)}$
    $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
    $\mathbf{D}_t^{(j)} \leftarrow \mathrm{MuonLMO}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$ ▷ LMO on the client
    if $\mathcal{C}^{\uparrow} = \mathrm{EF21}$ **then** ▷ EF21-SignMuon
      $\Delta_t^{(j)} \leftarrow \mathbf{D}_t^{(j)} - \mathbf{d}_{t-1}^{(j)};\quad \alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{d}_t^{(j)} \leftarrow \mathbf{d}_{t-1}^{(j)} + \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$
      send $\big(\mathrm{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else ▷ SignMuon
      send $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}(\mathbf{D}_t^{(j)})$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \mathrm{EF21}$ **then**
    $\mathbf{d}_t \leftarrow \mathbf{d}_{t-1} + \frac{1}{N}\sum_j \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$ ▷ aggregate direction
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \mathbf{d}_t$
  else
    $\hat{\mathbf{s}}_t \leftarrow \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \hat{\mathbf{s}}_t$
  end if
end for

Algorithm 9: Federated MuonUSign / MuonSign / EF21-MuonUSign / EF21-MuonSign (server-side LMO)

Input: initial model $\mathbf{X}_0$ (and $\mathbf{W}_0 = \mathbf{X}_0$ if $\mathcal{C}^{\downarrow} = \text{EF21-P}$); clients N; rounds T; learning rate η_t; momentum μ; compressors $\mathcal{C}^{\uparrow} \in \{\mathrm{sign}, \mathrm{EF21}\}$, $\mathcal{C}^{\downarrow} \in \{\mathrm{exact}, \mathrm{sign}, \text{EF21-P}\}$ (Table 10)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{g}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{G}_0 \leftarrow 0$
for $t = 1$ to T **do**
  broadcast $\mathbf{V}_{t-1} \leftarrow \mathbf{W}_{t-1}$ if $\mathcal{C}^{\downarrow} = \text{EF21-P}$, else $\mathbf{X}_{t-1}$
  for $j = 1$ to N **in parallel do** ▷ client j
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{V}_{t-1};\, \xi_t^{(j)})$
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\, \mathbf{G}_t^{(j)}$
    $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
    if $\mathcal{C}^{\uparrow} = \mathrm{EF21}$ **then** ▷ EF21-MuonUSign / EF21-MuonSign
      $\Delta_t^{(j)} \leftarrow \tilde{\mathbf{M}}_t^{(j)} - \mathbf{g}_{t-1}^{(j)};\quad \alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{g}_t^{(j)} \leftarrow \mathbf{g}_{t-1}^{(j)} + \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$
      send $\big(\mathrm{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else ▷ MuonUSign / MuonSign
      send $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \mathrm{EF21}$ **then**
    $\mathbf{G}_t \leftarrow \mathbf{G}_{t-1} + \frac{1}{N}\sum_j \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$ ▷ reconstruct gradient
  else
    $\mathbf{G}_t \leftarrow \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
  end if
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{G}_t)$ ▷ single LMO on the server
  if $\mathcal{C}^{\downarrow} = \mathrm{sign}$ **then** ▷ MuonSign
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \mathrm{sign}(\mathbf{D}_t)$; **broadcast** $\mathrm{sign}(\mathbf{D}_t)$
  else if $\mathcal{C}^{\downarrow} = \text{EF21-P}$ **then** ▷ EF21-MuonSign
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$
    $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1};\quad \alpha_t^{\downarrow} \leftarrow \mathrm{mean}(|\Delta_t^{\downarrow}|)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow}\, \mathrm{sign}(\Delta_t^{\downarrow})$; **broadcast** $\big(\mathrm{sign}(\Delta_t^{\downarrow}), \alpha_t^{\downarrow}\big)$
  else ▷ exact downlink: MuonUSign / EF21-MuonUSign
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$; **broadcast** $\mathbf{X}_t$
  end if
end for